
Need-Sufficiency Architecture and Gap Diagnostics for the B-to-C Gap

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Abstract

The companion paper [Lasser, 2026d, Revealed Sacrifice] proves the *Aggregate B-to-C Lower Bound* theorem for a privacy-respecting observation channel on realized capability volume vol_R , under admissibility conditions including S1 (free choice with non-degrading alternatives). This paper supplies the structural condition that makes S1’s below-sufficiency failure mode diagnosable (the actual filtering remains an institutional-attestation step; the condition identifies which events to filter, not by itself a filtering computation), together with the diagnostic architecture that interprets the bound’s output.

The *need-sufficiency architecture* is a poset-construction precondition distinguishing involuntary need-satisfaction from voluntary want-pursuit via multi-dimensional sufficiency thresholds and downstream-safe gates. We establish a sufficient condition under which retention degrades the agent below threshold and no third non-degrading alternative exists (S1 fails), filtering such events from the aggregate. A parallel *need-access cost* diagnostic tracks below-sufficiency expenditure as a companion metric.

The *gap decomposition* partitions the B-to-C gap into five cells (restricted, covered, dormant, residual, boundary-residual) under the default literal-partition variant. Cell classification is computable in near-linear $\tilde{O}(|P| + E + M)$ time given external admissibility labels and governance inputs. A *wireheading-consistent concentration signal* from the trade-flow Herfindahl–Hirschman Index is Schur-convex in the majorization order; the Part-A category-flow variant is third-party observable from public committed events plus the public S1-admissibility labelling. The *residual class* is re-characterized from a measurement-apparatus question to a channel-reach question. A worked example demonstrates the architecture end-to-end.

1 Introduction

The companion paper [Lasser, 2026d, Revealed Sacrifice] proves the *Aggregate B-to-C Lower Bound* theorem for the GFM framework’s exercised capability volume vol_R :

$$\text{vol}_R^{U,[W]} \geq \sum_n \Delta \text{vol}_P(X_n),$$

where the sum runs over admissible trade events in a window W whose bundles are pairwise poset-independent. The theorem converts the otherwise-unbounded B-to-C gap (the divergence between vol_P capability possession and vol_R capability exercise) into a bounded one on the subset of capability-space covered by observed sacrifice.

The theorem’s admissibility conditions are load-bearing. Part A requires S0 (valuation bounded by exercise), S1 (free choice with *non-degrading alternatives*), S2 (benchmark grounding), S4(a) (valuation additivity), and the genuine-exchange conditions of the revealed-sacrifice event definition.

Part B additionally requires S3, S5, and within-bundle poset-independence. Of these, S1 is the most empirically delicate: not every observed sacrifice is voluntary. An agent paying above market rate for medicine because they are in severe pain, paying above market rate for water because they are dehydrated, or paying above market rate for shelter because they are exposed are not voluntary trades in the S1 sense. Such sacrifices have *reversed polarity*: more sacrifice means the agent is in a *worse* situation, not revealing stronger preference.

This paper contributes the structural architecture that supplies a sufficient filter condition for one below-sufficiency failure mode of S1 (Definition 6): events in which retention degrades the agent and no third non-degrading alternative exists fail S1 and are filtered from the aggregate. Below-sufficiency *targeting* where a third non-degrading alternative is available is not by itself an S1 failure mode (S1 may still hold via that third alternative), but classification requires institutional attestation to confirm the third alternative’s existence and non-degrading status. Other S1 failure modes proper (coercion, duress, information asymmetry) likewise require institutional attestation rather than this structural filter (Remark 8). The paper also develops the diagnostic layers that interpret what the lower bound is signalling when a low β^{lower} fires an alarm.

1.1 What this paper does

1. **Need-sufficiency architecture (Section 2).** A poset-construction precondition distinguishing needs from wants via multi-dimensional sufficiency thresholds. The sufficiency threshold is the structural *polarity boundary*: it identifies the region in which the H1+H2 below-sufficiency S1-failure mode operates (retention degrades the agent below threshold and no third non-degrading alternative exists), and the region in which the structural failure mode does not apply, leaving the remaining S1 conditions (coercion, duress, information asymmetry, etc.) to institutional attestation. Events the attestation layer admits as S1-valid enter the companion paper’s aggregate [Lasser, 2026d, Revealed Sacrifice]; below-sufficiency targeting with a third non-degrading alternative likewise routes through attestation to determine S1-admissibility. A polarity-correct benchmark inverts cost dimensions; a multiplicative downstream-gate on vol_P -aggregation makes need-satisfaction a structural precondition for registering downstream weight; a parallel *need-access cost* diagnostic tracks the benchmarked capability surrendered for need-satisfaction.
2. **Gap decomposition (Section 3).** A five-cell partition of the B-to-C gap—restricted, covered, dormant, residual, boundary-residual—with an exact identity $1 - \beta^{\text{lower}} = \sum \delta$ under the default literal-partition variant (strategy (c)), preserving seed-cell attribution so that δ_{covered} remains interpretable as proxy-failure on observed capabilities. A four-cell variant under cooperative-closure semantics (strategies (a)/(b)) is also available, at the cost of priority-extension absorbing dormant and residual atoms into the covered cell. Supplies a near-linear $\tilde{O}(|P| + E + M)$ computability proposition from observed trade data plus governance-supplied structural inputs. Interprets alarm firings: low β^{lower} with high δ_{covered} deficit is proxy failure on observed capabilities; low β^{lower} with high $\delta_{\text{restricted}}$ or δ_{residual} reflects structural invisibility rather than proxy failure.
3. **Wireheading-consistent concentration signal (Section 4).** The Herfindahl–Hirschman Index on trade-flow leverage provides a tractable, governance-thresholded concentration alarm that is Schur-convex in the majorization order. A Part-A category-flow variant is third-party observable from public committed events plus the public S1-admissibility labelling.
4. **Residual class characterization (Section 5).** A revealed-observation curve of the residual class: the set of capabilities structurally outside the sacrifice channel’s reach. This differs from the classical benchmarkability-based residual class; an explicit witness shows $R_B \setminus R_S$ is non-empty, while $R_S \setminus R_B$ and $R_B \cap R_S$ depend on individuation choices (Proposition 8).
5. **Worked example (Section 6).** A two-substrate (biology + silicon) collective traced through four phases—healthy trade, automation-driven trade-flow concentration, diagnosis and recovery, and complementary- diagnostic cross-checking—demonstrates the full architecture end-to-end.

1.2 Relationship to the companion paper

The companion paper [Lasser, 2026d, Revealed Sacrifice] establishes the measurement theorem; this paper supplies the filter that keeps the channel clean and the diagnostic architecture that interprets its output. The division of labor:

Companion paper ([Lasser, 2026d, Revealed Sacrifice])	This paper
Sacrifice observation model, S0–S5	Need-sufficiency architecture (S1 filter)
Aggregate B-to-C lower bound	Gap decomposition
Commitment-layer composition	Wireheading-consistent alarm
Two sacrifice channels	Residual class characterization
Non-self-balancing finding	Worked example

We refer to the companion paper’s definitions (revealed-sacrifice event, assumptions S0–S5, the aggregate B-to-C lower bound theorem, committed event, privacy theorem) without re-deriving them; the companion paper’s notation conventions carry over. *Submission-package note.* This paper is intended for submission as a companion in the same package as [Lasser, 2026d, Revealed Sacrifice]; reviewers should have access to both. Where a venue requires a single self-contained submission, the load-bearing definitions (revealed-sacrifice event, S0–S5, aggregate lower bound) would be in-lined into a unified single-paper version. Stable archival identifiers will accompany the sequence as it reaches archival status.

1.3 Dependencies on prior papers

This paper inherits the GFM framework [Lasser, 2026b,f,c,g,a,e,h] and the measurement-theoretic infrastructure of the companion paper [Lasser, 2026d, Revealed Sacrifice]. A summary dependency table is collected in Appendix B.

2 Need-Sufficiency Architecture

The revealed-sacrifice observation channel ([Lasser, 2026d, Revealed Sacrifice]) admits only sacrifices satisfying S1 (free choice with non-degrading alternatives) into its aggregate, and on those S1-admissible events the channel reads sacrifice with positive polarity: more sacrifice equals more revealed value. S1 is the gate; the present paper supplies the structural condition that makes one of S1’s failure modes diagnosable. Without that condition, events that are structurally involuntary on a need dimension would, if mistakenly admitted, be misread with positive polarity when their actual polarity is reversed. An agent who pays \$500 for insulin is not revealing that insulin is worth \$500 beyond baseline, they are revealing that baseline functionality costs \$500 to maintain. The sacrifice magnitude reveals *cost-of-access* (how expensive it is for this agent to reach minimum viable functionality), not *preference strength* (how much the agent values the outcome beyond baseline). Larger sacrifice means a worse situation, not higher revealed value.

The framework needs a structural requirement on poset construction that distinguishes need-satisfaction from want-pursuit and identifies the boundary where sacrifice polarity flips.

2.1 Three motivating observations

Needs are neither purely maximizing nor minimizing. Every need has *both* maximizing dimensions (quality, quantity, reliability) and minimizing dimensions (cost, distance, time). Needs are not a single-polarity class. What *is* true: the minimizing dimensions of needs are where the polarity problem lives, because cost-of-access is involuntary.

Scalar sufficiency hides structural defects. A single “water access: sufficient” measurement might hide that the water is contaminated. The agent has water (passes the scalar threshold) but the way the need is met actively degrades future vol_P through downstream effects on health, cognition, and reproduction. The framework reports green while the system corrodes from the root.

Downstream-safe bundles. Three levels of sufficiency, each strictly stronger:

- *Level 1: Scalar threshold (insufficient).* “Water: yes/no.” Hides quality, reliability, and cost defects.
- *Level 2: Bundle threshold (necessary but not sufficient).* All dimensions of the need must independently meet their thresholds simultaneously. This catches contamination (quality dimension fails), but misses subtler failures where the satisfaction pathway degrades downstream capabilities over time. For example, when the contamination thresholds are incorrectly tuned.
- *Level 3: Downstream-safe bundle.* Bundle thresholds are all met *and* the need-satisfaction pathway does not produce net-negative effects on capabilities in the need’s downstream capability cone.

2.2 Formal definitions

Definition 1 (Multi-Dimensional Need Bundle). *A capability N identified as a need at position p_N in the poset has:*

- **Dimensions** $\{d_1, \dots, d_m\}$ with sufficiency thresholds $\{s_1, \dots, s_m\}$.
- Each dimension has a **polarity**: maximizing (quality, quantity, reliability) or minimizing (cost, distance, time).
- **Downstream cone** $\text{Down}(p_N) = \{c : c \text{ depends on } N \text{ in the poset}\}$.
- **Satisfaction pathway** π_N : the specific way the agent meets the need (which source, at what quality, at what cost).

For cost dimensions (minimizing polarity), the benchmark inverts via a domain-supplied positive cost reference $\text{cost}_k^{\text{ref}} > 0$: define $\text{actual}_k = \text{cost}_k^{\text{ref}} / \text{cost}_k$ and the sufficiency threshold $s_k = \text{cost}_k^{\text{ref}} / \text{cost}_k^{\text{max}}$ where $\text{cost}_k^{\text{max}}$ is the maximum acceptable cost. Both quantities are dimensionless ratios sharing the same cost-reference normalization, so actual_k and s_k live on a common scale and benchmark maximization aligns with cost minimization. The inversion is defined on $\text{cost}_k > 0$; the zero-cost limit is read as “no cost-dimension constraint binds” and treated as benchmark-saturated, with the convention $\text{actual}_k = \text{cost}_k^{\text{ref}} / \text{cost}_k$ capped at some domain-set ceiling $M_k > s_k$ to avoid the $\text{cost}_k \rightarrow 0^+$ singularity. The reference and ceiling are domain parameters; the paper does not legislate specific values.

Definition 2 (Downstream-Safe Sufficiency). *Let $\Pi_N^0 = \{\pi : d_k(\pi) = s_k \text{ for all } k\}$ be the class of pathways satisfying each dimension of N at exactly its sufficiency threshold, assumed non-empty by governance realizability hygiene. In discrete-valued domains where exact threshold-achievement may be infeasible, Π_N^0 is relaxed to pathways within a domain-set tolerance of the threshold; the framework’s downstream arguments are preserved by treating the relaxed set as the admissible reference base. Let $\Pi_N^{\text{ref}} \subseteq \Pi_N^0$ be a non-empty admissible reference set, designated externally by governance (Remark 1): the threshold-level pathways against which downstream non-degradation is evaluated. Need N at position p_N is downstream-safe sufficient under satisfaction pathway π_N iff:*

- **Bundle thresholds met:** $d_k(\pi_N) \geq s_k$ for all dimensions k (with cost dimensions inverted per Definition 1).
- **Downstream non-degradation.** For each downstream capability $c \in \text{Down}(p_N)$, let $\text{quality}_c : \Pi_N^0 \cup \{\pi_N\} \rightarrow \mathbb{R}$ be a domain-supplied scalar quality function on satisfaction pathways, providing a total order on pathways relative to capability c (e.g., for medical care, the standard-of-care quality function on care bundles; for water access, the contamination-adjusted access function). The framework reads quality_c as a governance input, not a computed quantity; the choice of quality function is part of the institutional adjudication that Definition 2 delegates to governance. Write $\text{quality}_c[\pi]$ for the quality of capability c when need N is satisfied via pathway π . Then π_N must dominate every $\pi^* \in \Pi_N^{\text{ref}}$ on every downstream capability simultaneously:

$$\text{quality}_c[\pi_N] \geq \text{quality}_c[\pi^*] \quad \text{for all } c \in \text{Down}(p_N), \pi^* \in \Pi_N^{\text{ref}}.$$

The universal quantifier over Π_N^{ref} is load-bearing: a merely existential condition (“dominate some threshold-level pathway”) would admit materially degrading pathways that happen to beat a pathologically weak reference. Under the common singleton case $\Pi_N^{\text{ref}} = \{\pi_N^{\text{ref}}\}$, the condition reduces to dominance against one canonical baseline. Each element of Π_N^{ref} must be a realizable pathway, not a coordinatewise supremum over Π_N^0 that no single pathway achieves.

Remark 1 (Governance designates the admissible reference set). *The admissible reference set Π_N^{ref} is a public governance input, not a framework-computed object. Institutions typically adopt a specific reference pathway as the canonical baseline (“standard of care” in medicine, “best available technology” in environmental policy); in those settings Π_N^{ref} is a singleton. Multi-reference settings arise when different disciplines canonicalize differently and π_N must satisfy all of them simultaneously; the downstream-safety condition then requires robust dominance over every admissible baseline. Governance may update Π_N^{ref} over time as standards evolve; the framework reads the current Π_N^{ref} at evaluation time and does not attempt to re-derive it from first principles.*

Remark 2 (Structural connection to exercise-pathway machinery). *The need-satisfaction pathway π_N is structurally an exercise pathway in the sense of [Lasser, 2026c, Definition 2]: a sequence of capability invocations producing an outcome (here, the satisfaction of need N). This identification lets the framework treat π_N as a first-class object rather than an abstract “way of satisfying the need.”*

The downstream-safety condition (Definition 2(b)) is, however, a distinct mathematical object from the concentration-risk measure of [Lasser, 2026c, Definition 3]. Concentration risk is a scalar worst-case contraction estimate over an entire pathway; downstream-safety is a per-capability quality comparison $\text{quality}_c(\pi_N) \geq \text{quality}_c(\pi^)$ against every reference pathway $\pi^* \in \Pi_N^{\text{ref}}$, indexed by $c \in \text{Down}(p_N)$. Neither reduces to the other. Evaluating $\text{quality}_c(\pi_N)$ in practice requires either (i) direct observation of c ’s performance conditional on the pathway, or (ii) structural inference from the poset’s cooperative structure. A full formalisation of the evaluation procedure is outside this paper’s scope; within the paper, Definition 2(b) is a structural condition the framework enforces (a pathway failing the condition is rejected as a satisfaction pathway for N), not a quantity the framework computes directly.*

Coherence with the framework’s privacy stance. Option (i) above (direct observation of c ’s performance conditional on the pathway) is precisely the kind of agent-interior measurement that the companion paper’s privacy commitments rule out at scale on the sacrifice channel. Downstream-safety evaluation therefore lives in one of two regimes: (a) public structural evaluation, where quality_c is determined from governance-supplied reference-pathway specifications and the poset’s structural cooperative/subsumption topology (no agent-interior access required, suitable for population-scale deployment); (b) consented-perimeter evaluation, where institutions with explicit consent (the regime of the companion paper’s exercise-indicator machinery, [Lasser, 2026d, Revealed Sacrifice, Appendix A]) directly observe pathway performance on enumerable downstream capabilities. Mixed deployments use (a) for poset-wide screening and (b) for adjudication of contested pathways within consented sub-domains. The architecture is coherent with the framework’s privacy stance under either regime: regime (a) requires no privacy commitment because no interior data is read, and regime (b) operates inside the perimeter where privacy commitments are explicitly waived. Outside both regimes (ad hoc agent-interior measurement at population scale), downstream-safety evaluation would be both infeasible and inconsistent with the framework’s stated deployment constraints, and the architecture should not be deployed in such a regime.

A policy consequence, not a theorem: because need-satisfaction pathways are not optional (the agent must satisfy the need), the framework’s response to a pathway failing Definition 2(b) differs from its response to a risk claim under [Lasser, 2026h, Controlled Relaxation]. A failing risk pathway is restricted (the capability is blocked). A failing satisfaction pathway must be replaced (the need must still be met, but through a pathway satisfying Definition 2). This governance distinction is upstream of the paper’s formal machinery; it names the correct operational response when the condition fails, not a new computation the framework performs.

Definition 3 (Sufficiency Threshold as Polarity Boundary). *The sufficiency bundle is the structural boundary where revealed-sacrifice polarity flips, conditional on the choice-set hypothesis stated below:*

- **Below sufficiency on any dimension, with no third non-degrading alternative:** The agent is investing to reach minimum viable functionality and has no alternative path that avoids

degrading some need. Sacrifice is involuntary (the agent must reach sufficiency to operate and has no other option). Sacrifice magnitude reveals **cost-of-access**: how expensive it is for this agent to reach baseline on this dimension. Larger sacrifice = worse situation.

- **Above sufficiency on all dimensions:** The agent has reached functionality on every need dimension; the need-insufficiency failure mode of S1 is removed. Sacrifice reveals **preference strength** provided the trade is voluntary in the residual S1 sense (free of coercion, duress, information asymmetry, and the other material confounds of S1; see Remark 8). Above-sufficiency status alone does not establish S1 in full; it removes one structural failure mode while leaving others to institutional attestation. Larger sacrifice = higher revealed value, conditional on residual S1 holding.
- **Below sufficiency on some dimension, but a third non-degrading alternative exists:** polarity is not determined by sufficiency status alone. S1 can still hold via the third alternative; classification requires institutional review (Definition 6 and Remark 8).

The sufficiency threshold is where the need-insufficiency failure mode of Assumption S1 ([Lasser, 2026d, Revealed Sacrifice]) becomes structurally diagnosable: events with retention-degradation and no third non-degrading alternative fail S1 by Definition 6; above sufficiency, this specific failure mode does not apply (though S1 can still fail via other mechanisms; see Definition 6 and Remark 8).

Definition 4 (Polarity-Correct Benchmark). For each dimension d_k of a need capability:

- **Below sufficiency:** $\text{benchmark}_k = \min(\text{actual}_k/s_k, 1)$. Climbs from 0 toward 1 as the agent approaches sufficiency. Caps at 1. Maximizing this equals closing the sufficiency gap.
- **Above sufficiency:** $\text{benchmark}_k = 1 + \alpha_k \cdot \max((\text{actual}_k - s_k)/s_k, 0)$, where $\alpha_k < 1$ discounts above-sufficiency gains relative to below-sufficiency gains at the margin: the derivative of benchmark_k with respect to actual_k is $1/s_k$ below sufficiency (where the benchmark is $\min(\text{actual}_k/s_k, 1)$) and α_k/s_k above sufficiency, so the per-unit ratio of marginal contribution is $\alpha_k : 1$ in favour of closing the gap rather than enhancing beyond it. This is a marginal-rate claim, not an absolute-magnitude claim: because the above-sufficiency term is unbounded in actual_k , sufficiently large above-sufficiency gains can dominate the closed-gap benchmark in absolute terms. Bounding total above-sufficiency contribution requires either a domain-specific cap on actual_k/s_k or replacing the linear ramp with a saturating function (e.g., $\alpha_k \cdot \min(\cdot, K)$); this paper leaves the cap as an architecture parameter.

Definition 4 operates at the single-capability level: it determines the benchmark score for each dimension of a given capability. It does not, by itself, model the structural fact that need-satisfaction is a precondition for the exercise of downstream capabilities: an agent without water does not merely lose one bit of water-access capability, they lose access to the entire class of cooperative capabilities that require a hydrated body. To capture this, the framework imposes a multiplicative gate on volp-aggregation through the downstream cone.

Definition 5 (Downstream-Gated Capability Weighting). For each capability $c \in P$, let $\mathcal{N}(c) = \{N : c \in \text{Down}(p_N)\}$ be the set of needs whose downstream cones contain c . For each such N , let $s_N(c) \in [0, 1]$ be the sufficiency level of N in the context where c is exercised, with $s_N(c) = 1$ when N is downstream-safe sufficient (Definition 2) and $s_N(c) = 0$ when N is fully unmet. In the common case where sufficiency is uniform across the downstream cone, we abbreviate $s_N(c) \equiv s_N$ and treat s_N as a scalar.

The effective weight of c is:

$$w^{\text{eff}}(c) = w(c) \cdot \prod_{N \in \mathcal{N}(c)} g(s_N(c)),$$

where $g : [0, 1] \rightarrow [0, 1]$ is a gate function satisfying: $g(1) = 1$, $g(0) = \alpha_0 \in [0, 1]$ (the gate's floor), g monotone non-decreasing, and g continuous on $(0, 1]$ with right-limit $\lim_{s \rightarrow 0^+} g(s) = \alpha_0$. Continuity on $(0, 1]$ excludes step-function gates with jump discontinuities in the interior; the right-limit condition at 0 pins the floor. Right-continuity alone (e.g., a step from α_0 at s_0 defined-from-the-right) would be insufficient because right-continuous step functions admit interior jumps; we

use the stronger continuous-on- $(0, 1]$ condition to rule them out. A gate with $\alpha_0 = 0$ is called a zero-floor gate; a gate with $\alpha_0 > 0$ is called a positive-floor gate. The four candidate gates of Remark 3 all satisfy continuity-on- $(0, 1]$ with the prescribed right-limit at zero; gates with interior jump discontinuities (e.g., bare step functions) are excluded. Linear, ReLU, and power-law gates are zero-floor; leaky-ReLU with parameter $\alpha > 0$ is positive-floor with $\alpha_0 = \alpha$.

The effective capability volume is:

$$\text{vol}_P^{\text{eff}}(g) = \sum_{c \in P} w^{\text{eff}}(c).$$

The summation is over the same capability census P that defines vol_P in [Lasser, 2026f, Proposition 1]; $\text{vol}_P^{\text{eff}}$ is constructed as a downstream-gated reweighting of vol_P 's atomic capability sum, with each $c \in P$ contributing $w^{\text{eff}}(c) = w(c) \cdot \prod_{N \in \mathcal{N}(c)} g(s_N(c))$ in place of $w(c)$. We treat $\text{vol}_P^{\text{eff}}$ as a gated diagnostic quantity for need-context optimization, not as a new poset-measure inheriting M1–M6 by construction. M5 non-negativity is preserved trivially (each gate factor is in $[0, 1]$, so $w^{\text{eff}}(c) \geq 0$). Compatibility with M4-type additivity on poset-independent direct sums and with M6-type cooperative superadditivity depends on whether the gate factors $\prod_N g(s_N(c))$ are constant across the relevant subset (so the gated weights factor cleanly); for homogeneous-sufficiency settings (Section 2.5) the factors are constant on each downstream cone and the additivity carries through, but the general case requires care and we do not claim $\text{vol}_P^{\text{eff}}$ inherits M4 or M6 unconditionally. The framework optimizes $\text{vol}_P^{\text{eff}}$ rather than raw vol_P once needs have been identified, treating the gated quantity as an objective-substitute justified by its incentive properties (Proposition 1, Proposition 2) rather than by axiom inheritance.

Remark 3 (Candidate gate functions). The choice of g is a design parameter that trades off smoothness, computational simplicity, and the strength of need-dominance. Four candidates:

- **Linear:** $g(s) = s$. Simplest; gentle need-dominance; $s_N(c) = 0.5$ halves the downstream contribution.
- **Leaky-ReLU:** $g(s) = \alpha$ for $s \leq s_0$, then linear ramp from α at s_0 to 1 at $s = 1$. Parameters: threshold $s_0 \in [0, 1)$, leak $\alpha \in [0, 1)$. Below threshold, downstream capabilities retain a constant floor α ; above, rapid ramp to full contribution.
- **ReLU:** Leaky-ReLU with $\alpha = 0$. Below threshold, downstream capabilities are fully suppressed; above, linear ramp. Hardest gate; strongest need-dominance.
- **Power law:** $g(s) = s^\gamma$ for $\gamma \geq 1$. Smooth everywhere; gradient vanishes at $s = 0$ for $\gamma > 1$, creating an optimization dead zone near zero sufficiency: an agent with severely unmet need would receive near-zero marginal incentive from $\text{vol}_P^{\text{eff}}$ to improve sufficiency, despite the situation being where the framework most wants improvement. This is a structural trade-off of high- γ power-law gates: severity is weighted strongly in absolute level (collapse magnitude is large) but weakly in marginal gradient (rate of improvement is small at low s); domains where this dead zone is unacceptable should use leaky-ReLU or low- γ power-law gates. Power law also gives progressively stronger penalty as the sufficiency gap widens. Single tunable parameter.

The appropriate gate shape depends on the domain. Biological needs with cliff-like failure modes (dehydration, oxygen deprivation) warrant ReLU or high- γ power law. Needs with graceful degradation (social connection, mild nutritional deficit) warrant leaky-ReLU or low- γ power law. The paper leaves g as a domain-specific design parameter and states subsequent results for any valid g .

Proposition 1 (Near-Total Collapse Under Unmet Need). Let g be any valid gate function (Definition 5) with floor $\alpha_0 = g(0) \in [0, 1)$, and suppose $s_N(c) \equiv s_N$ for all $c \in \text{Down}(p_N)$ (homogeneous sufficiency). Partition the poset as $P = \text{Down}(p_N) \sqcup \overline{\text{Down}(p_N)}$, with weight sums $W_{\text{down}} = \sum_{c \in \text{Down}(p_N)} w(c)$ and $W_{\text{out}} = \sum_{c \in \overline{\text{Down}(p_N)}} w(c)$, so $\text{vol}_P^{\text{raw}} = W_{\text{down}} + W_{\text{out}}$. If the agent is at sufficiency level $s_N < 1$:

$$\text{vol}_P^{\text{eff}}(g) \leq g(s_N) \cdot W_{\text{down}} + W_{\text{out}}.$$

Collapse limits (upper-bound form). As $s_N \rightarrow 0$, $g(s_N) \rightarrow \alpha_0$ by the floor condition. The proposition's upper bound becomes:

- Zero-floor gate ($\alpha_0 = 0$): the upper bound on the downstream contribution vanishes, $\text{vol}_P^{\text{eff}} \leq W_{\text{out}}$ in the limit. When $W_{\text{down}} \gg W_{\text{out}}$, the upper bound on effective volume falls to a small fraction of the unrestricted value (and the actual effective volume is at most this).
- Positive-floor gate ($\alpha_0 > 0$): the upper bound on downstream contribution falls to $\text{vol}_P^{\text{eff}} \leq \alpha_0 W_{\text{down}} + W_{\text{out}}$. The effective volume retains at most a fraction α_0 of downstream weight even under fully unmet need; equality with $\alpha_0 W_{\text{down}} + W_{\text{out}}$ requires all other gates evaluating to 1 (otherwise the additional gate factors push the actual effective volume strictly below the upper bound).

The multiplicative structure makes need-satisfaction a precondition for vol_P to register the full downstream portion of the agent's capabilities; an agent cannot maximize $\text{vol}_P^{\text{eff}}$ without closing sufficiency gaps on needs whose downstream cones cover their capability-weight. Whether unmet need collapses the downstream contribution entirely or only to the floor $\alpha_0 W_{\text{down}}$ is a design choice governed by the gate's floor parameter.

Proof. Split the sum over P by membership in $\text{Down}(p_N)$. For $c \in \text{Down}(p_N)$, the effective weight is $w^{\text{eff}}(c) = w(c) \cdot g(s_N) \cdot \prod_{N' \in \mathcal{N}(c), N' \neq N} g(s_{N'}(c)) \leq w(c) \cdot g(s_N)$ since each additional gate factor is ≤ 1 . For $c \in \overline{\text{Down}(p_N)}$, $N \notin \mathcal{N}(c)$, so $w^{\text{eff}}(c) = w(c) \cdot \prod_{N' \in \mathcal{N}(c)} g(s_{N'}(c)) \leq w(c)$. Summing the two cases:

$$\text{vol}_P^{\text{eff}} \leq g(s_N) \cdot \sum_{c \in \text{Down}(p_N)} w(c) + \sum_{c \in \overline{\text{Down}(p_N)}} w(c) = g(s_N) \cdot W_{\text{down}} + W_{\text{out}}.$$

Right-continuity of g at 0 gives $\lim_{s_N \rightarrow 0^+} g(s_N) = g(0) = \alpha_0$, so the downstream term tends to $\alpha_0 W_{\text{down}}$: zero for zero-floor gates, the floor value for positive-floor gates. ■

Remark 4 (The non-downstream floor and the gate floor). Two floors appear at $s_N = 0$: the always-present non-downstream floor W_{out} (capabilities not depending on N , which no gate can suppress), and the gate-dependent downstream floor $\alpha_0 W_{\text{down}}$ (which is zero for zero-floor gates and $\alpha \cdot W_{\text{down}}$ for leaky-ReLU with leak $\alpha > 0$). The worked table in Remark 6 exhibits both: at $s_N = 0$, linear, ReLU, and power-law gates give $0.231 = W_{\text{out}}/\text{vol}_P^{\text{raw}} = 3/13$ (only the non-downstream floor), while leaky-ReLU with $\alpha = 0.1$ gives $0.308 = (3+10 \cdot 0.1)/13 = (W_{\text{out}} + \alpha_0 W_{\text{down}})/\text{vol}_P^{\text{raw}}$ (both floors). The proposition's "total collapse" force applies to zero-floor gates with $W_{\text{down}} \gg W_{\text{out}}$; for positive-floor gates, the collapse is bounded below by the gate's floor regardless of $W_{\text{down}}/W_{\text{out}}$ ratio.

Proposition 2 (Need-Dominance Marginal Rate, conditional on a domain-supplied intervention metric). Assume homogeneous sufficiency $s_N(c) \equiv s_N$ for all $c \in \text{Down}(p_N)$ and that all other needs are at sufficiency ($s_{N'}(c) = 1$ for $N' \neq N$), so that s_N is a scalar. We compute partial derivatives of $\text{vol}_P^{\text{eff}}$ with respect to the design parameters s_N and $w(c')$. The dominance condition below is conditional on a domain-supplied intervention metric that prices unit increments of s_N and unit increments of $w(c')$ on a common cost scale; without that metric, the partial derivatives are not directly comparable and no formal dominance result follows. This is a design-convention statement, not a free-standing inequality on the framework's primitives. For any valid gate function g differentiable at s_N with $g'(s_N) > 0$, the two relevant partial derivatives of $\text{vol}_P^{\text{eff}}$ are

$$\frac{\partial \text{vol}_P^{\text{eff}}}{\partial s_N} = g'(s_N) \cdot W_{\text{down}}(N), \quad \frac{\partial \text{vol}_P^{\text{eff}}}{\partial w(c')} = \begin{cases} 1 & c' \notin \text{Down}(p_N), \\ g(s_N) \leq 1 & c' \in \text{Down}(p_N), \end{cases}$$

where $W_{\text{down}}(N) = \sum_{c \in \text{Down}(p_N)} w(c)$. The two derivatives have different domains (s_N is a dimensionless sufficiency level in $[0, 1]$, while $w(c')$ carries vol_P -weight units) and are not directly commensurable. A meaningful comparison requires a domain-supplied intervention-cost map $\kappa : \Delta s_N \times \Delta w \rightarrow \mathbb{R}_{\geq 0}$ that prices increments in s_N and increments in w on a common cost scale; the marginal-dominance discussion below is the statement of when, under such a κ identifying one unit of Δs_N with one unit of Δw as equally costly, the sufficiency gradient dominates. Without κ , no comparison is meaningful; see Remark 5.

Marginal-dominance condition (subject to a domain-supplied intervention metric). “Closing the sufficiency gap is more valuable than adding unit weight” is a statement about a per-unit-of-intervention comparison; it requires a domain-supplied metric that maps sufficiency increments and weight increments to a common intervention-cost scale. Under such a metric, with one unit of Δs_N and one unit of Δw identified as equally costly, the marginal return to closing the sufficiency gap on N exceeds the marginal return to unit-weight addition whenever

$$g'(s_N) \cdot W_{\text{down}}(N) > 1.$$

Without the metric, the partial derivatives are not directly comparable and no marginal-dominance claim follows from the formula alone. For linear gates ($g(s) = s$), this reduces to $W_{\text{down}}(N) > 1$. For leaky-ReLU or ReLU gates, the condition holds at any s_N in the ramp region provided $W_{\text{down}}(N)$ exceeds the reciprocal of the local slope. For power-law gates $g(s) = s^\gamma$ with $\gamma > 1$, the condition fails near $s_N = 0$ (where $g'(s_N)$ is small) even for large downstream cones, an intentional feature of gates that weight severity of unmet need over its breadth.

Proof. Under the stated hypothesis, every gate product collapses: for $c \in \text{Down}(p_N)$, the product $\prod_{N' \in \mathcal{N}(c)} g(s_{N'}(c))$ equals $g(s_N) \cdot \prod_{N' \neq N} g(1) = g(s_N)$. Thus $w^{\text{eff}}(c) = w(c) \cdot g(s_N)$ for $c \in \text{Down}(p_N)$ and $w^{\text{eff}}(c) = w(c)$ otherwise. Summing gives $\text{vol}_P^{\text{eff}} = g(s_N) \cdot W_{\text{down}}(N) + W_{\text{out}}(N)$. Only the first term depends on s_N , so differentiation yields $\partial \text{vol}_P^{\text{eff}} / \partial s_N = g'(s_N) \cdot W_{\text{down}}(N)$. Differentiating in $w(c')$: the capability c' 's gated weight is $g(s_N) \cdot w(c')$ if $c' \in \text{Down}(p_N)$ and $w(c')$ otherwise, giving the stated partial derivatives. Comparing the two partials on the common unit-step scale yields the dominance condition. ■

Remark 5 (Finite-step form and intervention-cost scale). *The marginal-dominance condition is a per-unit-cost comparison under the convention that one unit of Δs_N and one unit of Δw have the same intervention cost. Its finite-step form is the Taylor expansion: a sufficiency increment $\Delta s_N = \delta$ produces gain $[g(s_N + \delta) - g(s_N)] \cdot W_{\text{down}}(N) = g'(s_N)W_{\text{down}}(N) \cdot \delta + O(\delta^2)$, while a weight increment $\Delta w(c') = \delta$ on a single capability produces gain $\delta \cdot \max(1, g(s_N)) \leq \delta$. For small enough δ , the marginal condition $g'(s_N)W_{\text{down}}(N) > 1$ implies the sufficiency increment strictly dominates; a finite δ for which $[g(s_N + \delta) - g(s_N)] \cdot W_{\text{down}}(N) > \delta$ is the exact condition. The condition $g'(s_N)W_{\text{down}}(N) > 1$ is not a claim that a finite Δs_N of arbitrary size beats a finite unit of weight: e.g., with $g(s) = s$ and $W_{\text{down}} = 10$, a $\Delta s_N = 0.01$ gain of 0.1 does not exceed a unit-weight gain of 1 in absolute terms, only in per-unit-of-intervention terms (the sufficiency increment returned ten times the per-unit gain). Interpreting the condition as anything other than a marginal-rate comparison would conflate the two intervention scales.*

Remark 6 (Worked comparison across candidate gates). *Consider an agent with 10 cooperative capabilities in $\text{Down}(p_N)$ (weight 1 each) and 3 capabilities outside the cone (weight 1 each), so $\text{vol}_P^{\text{raw}} = 13$. At homogeneous sufficiency level $s_N(c) \equiv s$ for all $c \in \text{Down}(p_N)$, $\text{vol}_P^{\text{eff}}(s) = 3 + 10 \cdot g(s)$. The following table compares effective volume as a fraction of $\text{vol}_P^{\text{raw}}$ for the four candidate gates (Leaky-ReLU with $\alpha = 0.1$, $s_0 = 0.8$; ReLU with $s_0 = 0.8$; power law with $\gamma = 2$):*

s	Linear	Leaky-ReLU	ReLU	Power ($\gamma=2$)	Power ($\gamma=3$)
0.00	0.231	0.308	0.231	0.231	0.231
0.25	0.423	0.308	0.231	0.279	0.243
0.50	0.615	0.308	0.231	0.423	0.327
0.75	0.808	0.308	0.231	0.663	0.556
0.80	0.846	0.308	0.231	0.723	0.625
0.90	0.923	0.654	0.615	0.854	0.792
1.00	1.000	1.000	1.000	1.000	1.000

Linear gives the gentlest gate (at $s = 0.5$, downstream contributions are only halved, so 62% of raw vol_P is still registered). ReLU gives the hardest gate (below threshold $s_0 = 0.8$, downstream contributions are fully suppressed; $\text{vol}_P^{\text{eff}}$ stays at the non-downstream floor of $3/13 \approx 23\%$). Power-law gates sit between, with higher γ approaching ReLU's sharpness. Leaky-ReLU preserves the ReLU shape above threshold while leaving a small floor α below.

2.3 Sacrifice-polarity (S1) connection

Definition 6 (Below-sufficiency sacrifice). *Let $\mathcal{A}_i$ denote agent i 's choice set at trade time, including the trade (Y_n for X_n), the retention alternative (refuse the trade, keep X_n), and any other alternative actions or trades available. A sacrifice event (i, X_n, Y_n, t_n) is a below-sufficiency sacrifice when both:*

- (H1) *retention is degrading: keeping X_n leaves the agent below the sufficiency threshold s_k on some need dimension d_k ;*
- (H2) *no alternative in $\mathcal{A}_i$ other than the trade itself is non-degrading on every need dimension; that is, for every $a \in \mathcal{A}_i \setminus \{(the\ trade)\}$, there exists some need dimension $d_{\ell(a)}$ with threshold $s_{\ell(a)}$ such that taking action a leaves the agent below $s_{\ell(a)}$ on $d_{\ell(a)}$ (the failing dimension may differ across alternatives, and need not coincide with the d_k from H1).*

Remark 7 (Below-sufficiency sacrifices fail S1). *A below-sufficiency sacrifice fails S1 ([Lasser, 2026d, Revealed Sacrifice]) directly: H2 asserts that no alternative-to-the-trade in $\mathcal{A}_i$ is non-degrading, which is exactly the S1 failure condition (S1 requires the existence of at least one non-degrading alternative to the trade). H1 in this construction names why the alternatives are degrading (retention is one specific degrading option), ensuring the failure is need-driven rather than artifactual. The named-pattern value of Definition 6 is operational: it picks out the structural subclass of S1 failures that the need-sufficiency architecture identifies (retention is degrading and no third option escapes degradation), letting the institutional attestation layer filter such events without rederiving the S1 logic per case. Below-sufficiency targeting (sacrifice aimed at improving an unmet need dimension where H1 holds but H2 fails because $\mathcal{A}_i$ contains a non-degrading non-trade alternative) does not satisfy Definition 6 and does not imply S1 failure: S1 can still hold via that third alternative, in which case the trade reveals genuine preference between non-degrading options.*

Consequence for the sacrifice channel. Below-sufficiency sacrifices in the H1+H2 narrow sense of Definition 6 (retention degrades *and* no third non-degrading alternative exists) are filtered from the revealed-sacrifice aggregate by S1 before they reach the aggregate lower bound (the aggregate lower bound theorem [Lasser, 2026d, Revealed Sacrifice]). Below-sufficiency *targeting* where a third alternative exists is not automatically filtered by this definition: such events may still satisfy S1 if the agent's choice set contains a non-degrading option, and the institutional attestation layer must determine which class an event belongs to. For above-sufficiency events (retention does not put the agent below threshold), the need-insufficiency failure mode of S1 is removed structurally. S1 may still fail for above-sufficiency events via other mechanisms (coercion, duress; see Remark 8 below).

Remark 8 (S1 is not automatically satisfied above sufficiency). *S1 can still fail for above-sufficiency events if the trade was coerced by other mechanisms (economic duress, political pressure, information asymmetry) even though the need-sufficiency condition is met. The sufficiency threshold resolves one structural failure mode of S1 (the involuntary-need-satisfaction case); other S1 failure modes (economic duress, information asymmetry, and the coercion open question discussed in the companion paper [Lasser, 2026d, Revealed Sacrifice]) remain.*

Remark 9 (Operational sourcing of the H1+H2 conditions). *H1 (retention degrades) is checkable from the agent's sufficiency state on each need dimension, which is computable from the poset construction once thresholds are supplied. H2 (no third non-degrading alternative) is the harder claim: it quantifies over the agent's full choice set $\mathcal{A}_i$, which is not directly observable from public trade data. In practice, H2 must be established by the institutional attestation layer of the companion paper [Lasser, 2026d, Revealed Sacrifice] (the external-attestation remark there describes how per-event hypotheses about the agent's choice set are certified through trade-level audit or attestation, not through the public partition $\mathcal{C}$ alone). Without H2 attestation, an event with H1 holding should be classified as below-sufficiency targeting (potentially S1-admissible) rather than below-sufficiency sacrifice (S1-failing); the conservative default is to require explicit H2 attestation before filtering.*

2.4 The capability-stuffing problem

An agent optimizing vol_P under the non-triviality axiom M5 (every valid capability with $s_{\max} \geq 1$ contributes positive weight [Lasser, 2026f, Proposition 1]) has a natural incentive to add capabilities that pass structural validation but do not increase experiential optionality. We call this the *capability-stuffing* failure mode.

The canonical form is redundant capability construction targeting a single underlying need from multiple vantage points. “Travel from home to a gas station” is a real, benchmarkable *do*-capability: it satisfies the axioms M1–M6, has a well-defined metric ($s_{\max} \geq 1$), and represents genuine structural knowledge about the world. Adding “travel from work to a gas station,” “travel from the grocery store to a gas station,” and so on produces a sequence of further real, benchmarkable capabilities, each passing the same axiomatic checks. Collectively they inflate vol_P without broadening what the agent can actually do: every addition instantiates the same underlying optionality (fuel access) from a different origin.

Capability-stuffing is distinct from other failure modes in the sequence. Wireheading (Lasser, 2026a, Discussion) is narrow self-reinforcing exercise of existing capabilities, whereas stuffing adds new capabilities without exercising them degenerately. The Doll Problem (Lasser, 2026b, Section 7.1) concerns possessed-but-unexercised capabilities, whereas stuffed capabilities may be technically exercised (routing through the gas station “exercises” the distance-measurement). The B-to-C gap in the aggregate sense this paper otherwise treats is widened by stuffing through a specific mechanism (individual-weight inflation) that the aggregate lower bound theorem [Lasser, 2026d, Revealed Sacrifice] does not, by itself, address. Stuffing is a framework-native perverse instantiation: the optimizer discovers it by following the vol_P -maximization gradient under M5, not by departing from the framework’s rules.

Tension with structural discovery. Capability-stuffing is closely related to a *legitimate* mechanism: the structural discovery value of [Lasser, 2026c, Proposition 2], which holds that discovering new poset structure is value-positive. An agent adding a new distance-measurement capability to its repertoire is making a structural discovery in that sense. The formal distinction between legitimate discovery and capability-stuffing turns on whether the new capability creates new cooperative structure:

- **Legitimate structural discovery** enriches the poset by creating cooperative capabilities (axiom M6 superadditivity): the new capability interacts with existing capabilities to produce outputs previously unachievable. The marginal contribution to vol_P includes cooperative terms beyond the individual weight.
- **Capability-stuffing** adds individual capabilities whose marginal cooperative contribution is zero (or effectively so): the new capability’s weight accrues through M5 alone, not through M6. No new cooperative outputs become achievable.

Without a structural mechanism that distinguishes the two, the framework rewards both equally, and a vol_P -maximizer with capability-discovery latitude will exploit the undifferentiated reward to stuff.

What the need-sufficiency architecture does resolve. The downstream-safe bundle sufficiency condition (Definition 2) and the polarity-correct benchmark (Definition 4) together handle one specific class of redundant construction: capabilities whose claimed contribution lies *below* the sufficiency threshold on an already-satisfied need dimension. For those, the below-sufficiency benchmark saturates at 1, so additional instances do not inflate the benchmark. This is a real structural property of the architecture and it is worth stating narrowly:

Proposition 3 (Below-Sufficiency Saturation (pointwise)). *For any need dimension d_k of any single capability, the below-sufficiency component of the polarity-correct benchmark (Definition 4) saturates at 1 once $\text{actual}_k \geq s_k$. This is a pointwise cap: the benchmark value assigned to dimension d_k of one capability does not exceed 1, regardless of how many times that capability’s actual_k is re-evaluated or re-instantiated.*

Proof. Below-sufficiency benchmark for dimension d_k is $\min(\text{actual}_k/s_k, 1)$, which is bounded above by 1 pointwise. ■

Remark 10 (The pointwise cap does not extend to an aggregate dimension-level cap). *Proposition 3 is a per-capability, per-dimension cap, not a dimension-level aggregate cap. Distinct capabilities $c_1, c_2, \dots$ each reporting $\text{actual}_k \geq s_k$ on the same dimension d_k each score 1 under the below-sufficiency component; their M5-weighted sum in vol_P is not capped by this proposition and accumulates with each new capability. The proposition therefore blocks only a narrow class of stuffing: re-instantiating a single capability to inflate its own benchmark via repeated reporting. The*

broader class of capability-stuffing (creating distinct capabilities that each satisfy sufficiency on the same dimension) is not structurally blocked and is acknowledged as unresolved below.

What the need-sufficiency architecture does not resolve. Proposition 3 is the full extent of the architecture’s structural anti-stuffing contribution. It does *not* cover the following:

- (i) Above-sufficiency re-instantiation. An attacker can construct capabilities with distinct actual_k values exceeding s_k (e.g., home-to-gas at 15 minutes and work-to-gas at 5 minutes). Each earns positive above-sufficiency credit $\alpha_k(\text{actual}_k - s_k)/s_k$ under Definition 4, even though the underlying optionality (fuel access) is unchanged.
- (ii) Capabilities whose claimed dimension is not inside any existing need’s downstream cone, for which the gate function (Definition 5) does not apply and M5 weight accrues fully to vol_P .
- (iii) Arbitrary asymmetric decoration (distinct cooperative descendants, minor weight perturbations, unique subsumption edges) that preserves M5 credit for the stuffed capability regardless of whether the capability provides genuinely new optionality.

Each of these defeats any attempt to block stuffing using post-declaration poset properties or exercise patterns alone. The honest statement of the need-sufficiency architecture’s contribution is sacrifice-polarity resolution (via Definition 6 above, which is load-bearing for the sacrifice channel) together with the narrow below-sufficiency cap of Proposition 3. Full structural resolution of M5 stuffing requires machinery this paper does not develop.

Capability-stuffing as structural avoidance. The failure pattern across attempts is that every signature over post-declaration data is adversarially manipulable. The move that changes the framing is recognizing stuffing as a specific instance of *structural avoidance*, the pathology flagged as an open gap in [Lasser, 2026f, Discussion]: when the framework rewards capability declarations additively, an agent optimizing vol_P has an incentive to *decline to discover* that capabilities merge, factor, or are otherwise reducible. Stuffing is the concrete manifestation: the agent keeps “travel from home to gas station” and “travel from work to gas station” distinct in P rather than merging them into the canonical form “access to fuel from multiple origins” that a proper structural discovery would produce. Each unmerged capability inflates the M5 count; the merged capability has the same optionality content.

The positive dual is already in the sequence: [Lasser, 2026c, Proposition 2] establishes that structural discovery is value-positive under conditions on the discovered structure’s information value. Making the symmetric claim (that refusal to discover is value-negative) requires mechanism, not entailment. The direction is a *proof-burden taxonomy* with *claim-and-challenge back-pressure*.

The proof-burden direction. For each avoidance route, a corresponding burden the agent must discharge to retain vol_P credit. Three types: proof-of-non-mergeability (for the stuffing-specific case), proof-of-provenance (blocking declarations without underlying structural work), and proof-of-use (blocking declarations that remain persistently dormant outside the residual class R_S of §5). Proof-burdens operate as preprocessing on the poset before M5 is applied; the M5 formula is preserved, but the measure is evaluated on the canonicalized poset rather than the presented one. Canonicalization itself is new axiomatic content at the preprocessing layer.

Claims are temporal and contestable. An agent’s proof-of-non-mergeability stands until a counter-argument demonstrates a merger mechanism; the discoverer of that mechanism receives structural-discovery credit ([Lasser, 2026c, Proposition 2]); the avoider loses the M5 credit whose proof has been voided. This makes avoidance economically risky rather than economically stable. Analogous architectures exist in scientific consensus, patent challenge, and proof-of-stake slashing; adapting any of them requires substantive protocol and calibration work.

Scope of the follow-up. The sequence provides substrate for parts of this construction: The GFM commitment protocol for timestamped claims, the companion paper’s sacrifice channel [Lasser, 2026d, Revealed Sacrifice] for partial use-evidence, and risk controlled relaxation as an adjudication pattern. Substrate is not implementation: formalizing the canonicalization operator, the three proof-burden types with admissible evidence classes, the claim-and-challenge protocol with challenge-

market robustness (admission filtering against spam, identity weighting against Sybil flooding, independence requirements against cartel suppression), and the equilibrium-dominance theorem requires a dedicated follow-up paper. This paper states the direction and leaves the formal work there. The open-question entry in §8 describes the follow-up scope in more detail.

2.5 How sufficiency thresholds are set

The framework treats sufficiency thresholds as *governance-supplied inputs*: the formal apparatus reads $\{s_k\}$ at evaluation time and does not derive them from first principles within the paper’s measurement-theoretic machinery (consistent with the limitations statement of Section 8). What is structurally rigorous in the architecture is the *downstream-safety condition* (Definition 2) that any candidate π_N must satisfy at a given threshold, not the threshold-setting procedure itself. Four upstream mechanisms can produce the governance input, not mutually exclusive:

(a) Poset-structural. Capabilities with high fan-out (many downstream dependents) have their sufficiency level set by the minimum required for downstream capabilities to function. Computable from poset topology.

(b) Empirical / cross-sectional. Observe the population’s actual capability levels and identify the threshold below which agents demonstrably cease functioning on downstream capabilities.

(c) Revealed-sacrifice-derived (heuristic). Sufficiency is approximated by the level at which sacrifice polarity statistically flips in aggregate. As a *transition-region heuristic*, observed sacrifice trends toward involuntary, retention-degrading-and-no-alternative patterns below the threshold and toward voluntary, choice-driven patterns above it; the transition point in the sacrifice data approximates the threshold empirically. This is the polarity-classification statistic on aggregate behaviour, not the per-event S1 admissibility rule itself: per-event S1 classification follows Definition 6’s H1+H2 condition (which distinguishes structurally-diagnosable below-sufficiency S1 failure from below-sufficiency targeting with a third non-degrading alternative), and the revealed-sacrifice transition region serves as a calibration input rather than a substitute for that rule. This mechanism also has a bootstrap problem: identifying the threshold requires classifying events, but classifying events requires the threshold. This mechanism is heuristic, not theorem-derived: the revealed-sacrifice channel does not by itself certify a threshold value, only suggest a transition region for governance to ratify. Resolution in practice: start with structural/normative thresholds (mechanisms (a) and (d)), refine using sacrifice data.

(d) Normative / external. A human authority or governance process declares sufficiency levels. This is the approach of Sen [1999] and Nussbaum [2011]. The framework’s formal apparatus is agnostic to the source; it needs the thresholds as input.

Operational stance. Initial thresholds are set by (a) (poset-structural) and (d) (normative), which are non-bootstrapping. Once enough sacrifice events are observed, (c) (revealed-sacrifice-derived) refines the initial values empirically, with the polarity-transition region in the trade data calibrating the threshold against agents’ actual behavioural breakpoint. Mechanism (b) (cross-sectional empirical) plays a similar refinement role on the population-level functioning side. No single mechanism is “primary”; (a)+(d) initialize, (b)+(c) refine.

Descriptive vs. normative thresholds. Mechanisms (a)–(c) produce *descriptive* thresholds: they identify levels at which downstream functionality breaks down or sacrifice polarity inverts, which are empirical or structural facts about agents in the deployment context. Mechanism (d) produces a *normative* threshold: the level a human authority has *decided* agents should not have to fall below, irrespective of the descriptive breakdown level. Descriptive and normative thresholds can diverge in either direction: a society may set a normative food-sufficiency threshold above the biological minimum to respect dignity, or (perversely) a normative threshold below the descriptive breakdown level for budgetary reasons. The framework’s formal apparatus reads whatever threshold governance supplies and does not by itself adjudicate between descriptive and normative provenance; downstream-safety and the structural arguments of Section 2 proceed identically under either kind of threshold. However, the *interpretation* of δ_{covered} alarms and the meaning of below-sufficiency event filtering depend on which kind of threshold is in force: a poset-structural threshold means

“below this level, downstream capabilities literally cannot function”; a normative threshold means “below this level, the governance regime has stipulated agents should not have to sacrifice.” Deployments should document the provenance of each threshold so that diagnostic output is read against the correct interpretive backdrop. Mixed deployments (structural thresholds for biological needs and normative thresholds for socially-constructed needs) are operationally common, and the framework accommodates them as long as each threshold’s provenance is recorded with the threshold value.

3 Alarm Mechanism and Gap Decomposition

3.1 The alarm

Definition 7 (B-to-C Evidence Alarm). *The B-to-C evidence alarm fires when the lower-bound B-to-C ratio drops below a threshold:*

$$\text{ALARM}(T) = \beta^{\text{lower}}(P, T) < \beta_{\text{alarm}},$$

where $\beta^{\text{lower}}(P, T)$ is the global conservative ratio of the companion paper [Lasser, 2026d, Revealed Sacrifice] on the full capability poset P (numerator $\text{vol}_R^{\text{lower}}(U, T)$ extended by zero outside the observed subset U , denominator $\text{vol}_P(P)$), and $\beta_{\text{alarm}} \in (0, 1)$ is configurable. The gap-decomposition identity below is stated against this same denominator $\text{vol}_P(P)$; the alarm and decomposition use a common denominator throughout. The alarm is an evidence alarm, not a proxy-failure alarm directly: when β^{lower} is low, the sacrifice channel shows little evidence of vol_R -content proportional to vol_P , but this does not by itself imply that true exercise is low. The interpretation of a firing alarm requires additional analysis (via the gap decomposition below) to determine whether low evidence reflects genuine low exercise, restricted capabilities, dormant wants, or residual-class capabilities beyond the channel’s reach.

Remark 11 (Evidence alarm versus proxy-failure diagnosis). *Let $\beta_{\text{true}} := \text{vol}_R / \text{vol}_P$ be the true (latent) exercised fraction, of which $\beta^{\text{lower}} = \text{vol}_R^{\text{lower}} / \text{vol}_P$ is the sacrifice-channel lower bound. Because $\beta^{\text{lower}} \leq \beta_{\text{true}}$ by construction, the alarm distinguishes two regimes by which δ term dominates the gap $1 - \beta^{\text{lower}}$:*

- **Alarm fires and δ_{covered} dominates the gap:** *on the covered cell, observed trade evidence is small relative to that cell’s vol_P -weight ($\delta_{\text{covered}} = [\text{vol}_P(C) - \text{vol}_R^{\text{lower}}(C)] / \text{vol}_P(P)$ is large, indicating $\text{vol}_R^{\text{lower}}(C) \ll \text{vol}_P(C)$). This is the proxy-failure-candidate signature: capabilities the sacrifice channel can reach are present in vol_P but registering little exercise evidence. Reaching a proxy-failure conclusion from this signature additionally requires ruling out the lower-bound’s slack as the explanation: low $\text{vol}_R^{\text{lower}}(C)$ relative to $\text{vol}_P(C)$ can reflect (i) genuine low exercise (the proxy-failure reading), (ii) Part-A supremum or Part-B max-attribution loose against the true $\text{vol}_R(C)$ (the lower-bound is conservative by construction, and its conservatism varies with bundle structure and the $\alpha_{n,c}$ defaults), (iii) admissibility filtering rejecting events that would otherwise contribute to $\text{vol}_R^{\text{lower}}(C)$, or (iv) sparse observation in W . The covered-deficit signature is necessary but not sufficient for a proxy-failure determination; institutional review of the explanation is required before escalation.*
- **Alarm fires and δ_{residual} or $\delta_{\text{restricted}}$ dominates:** *the low β^{lower} reflects structural invisibility or policy restriction, not proxy failure. The framework should not escalate in these cases; the alarm is firing on what the sacrifice channel cannot reach, not on degraded exercise on what it can reach.*

As a diagnostic for genuine vol_R -degradation, the alarm has both false-positive and false-negative regimes:

- False positives (alarm fires, no proxy failure) arise when low β^{lower} reflects channel-blindness (high δ_{residual} or $\delta_{\text{restricted}}$), bound looseness (Part-A’s supremum or Part-B’s max-attribution falling well below the truth on covered capabilities), admissibility-filter exclusions, or sparse observation rather than degraded exercise. The gap decomposition’s cell-attribution is the diagnostic that distinguishes these from genuine proxy failure.

- False negatives (*alarm does not fire, but proxy-failure pathology is present*) arise when wireheading or other pathologies produce high $\text{vol}_R^{\text{lower}}$ on loop-driver categories without depressing β^{lower} below the threshold; the wireheading-consistent concentration signal of Section 4 is the diagnostic complementing β^{lower} for these cases. Pathologies that neither lower β^{lower} nor concentrate observed flow are outside the joint signal's reach.

Both regimes are structural costs of privacy-respecting observation; the multi-diagnostic combination (Proposition 5) is what reduces their individual blind spots.

3.2 Diagnostic decomposition

Definition 8 (Gap Decomposition Under Revealed Sacrifice). *Under the Part B hypotheses of the aggregate lower bound theorem of the companion paper [Lasser, 2026d, Revealed Sacrifice] (S0–S5), the per-capability aggregate $\text{vol}_R^{\text{lower}}(c)$ is well-defined, and subset aggregates $\text{vol}_R^{\text{lower}}(A)$ are given by the subset-extension of $\text{vol}_R^{\text{lower}}$ [Lasser, 2026d, Revealed Sacrifice]. Under those hypotheses, the B-to-C gap $(1 - \beta^{\text{lower}})$ decomposes into disjoint sources whose count depends on the closure strategy (Remark 13): five under the default literal-partition strategy (c), four under cooperative-closure strategies (a)/(b). Each individual capability in P is assigned to exactly one cell by priority ordering (restricted > covered > dormant > residual); under (c), boundary-crossing cooperative nodes form a fifth bucket B . Each cell's δ term is its uncovered vol_P share:*

$$1 - \beta^{\text{lower}} = \delta_{\text{restricted}} + \delta_{\text{covered}} + \delta_{\text{dormant}} + \delta_{\text{residual}} + \delta_{\text{boundary}},$$

where δ_{boundary} is omitted under strategies (a)/(b) (which absorb cross-cell cooperatives via priority-extension into a participant's cell rather than routing them to B). The capability partition is constructed in two stages. **Stage 1 (raw seed cells, by priority)**. Classify individual capabilities by the priority rule:

- **Seed restricted cell:** $R_0 := \Xi$, where Ξ is the set of capabilities currently restricted under the preemptive-restriction criterion of [Lasser, 2026c, Proposition 1]. Classified first regardless of sacrifice evidence.
- **Seed covered cell:** $C_0 := C_S \setminus \Xi$, where C_S is the set of capabilities appearing in at least one acquired bundle Y_n of a theorem-admissible sacrifice event in $[T - H, T]$. A theorem-admissible event for Part A satisfies S0, S1, S2, and S4(a); for Part B it additionally satisfies S3, S5, and within-bundle poset-independence (the event-local bundle decomposition [Lasser, 2026d, Revealed Sacrifice]).
- **Seed dormant cell:** $D_0 := D_S$, where $D_S = \{d \in P : d \notin \Xi, d \notin R_S, d \notin C_S\}$. Tradable-in-principle capabilities with zero want-pursuit sacrifice evidence (dormant wants).
- **Seed residual cell:** $S_0 := R_S \setminus \Xi$, where R_S is the structural residual class (Definition 12). Residual capabilities outside Ξ ; capabilities in $\Xi \cap R_S$ are already assigned to R_0 by priority.

The four seed cells $\{R_0, C_0, D_0, S_0\}$ are pairwise disjoint and their union is P : R_0 and S_0 are disjoint by construction; C_0 and D_0 are priority-filtered to exclude Ξ ; $C_0 \cap R_S = \emptyset$ because residual capabilities admit no sacrifice-observable path in principle (Definition 12) and therefore cannot appear in any admissible bundle; remaining pairs are disjoint by the bullet-stated exclusions.

Stage 2 (closure-adjusted cells). The seed cells may contain cooperative or subsumption edges crossing cell boundaries; to obtain a closed additive identity we apply a closure policy. The default is the literal-partition variant (strategy (c) of Remark 13), which preserves seed-cell attribution by routing every boundary-crossing cooperative node into a separate boundary-residual bucket B rather than absorbing it into a participant's cell; this yields the five-cell partition $\{R, C, D, S, B\}$ used by the worked example, with the additivity identity following from the strategy-(c) atomic-weight accounting convention $\text{vol}_P(X) := \sum_{c \in X} w(c)$ summed over the disjoint cells (no M4 invocation; see Remark 13 for the precise semantics). Under cooperative-closure semantics (strategies (a)/(b)), an alternative four-cell partition $\{R, C, D, S\}$ is available at the cost of priority-extension absorbing seed-dormant and seed-residual atoms into the covered cell when they are

poset-connected to a non-residual node. In either regime the cells are pairwise disjoint, union to P , and agree with the seeds on all individual-capability classifications except where closure redistributes poset-connected components (Remark 12 discusses when the absorption occurs under (a)/(b) and the condition under which the residual cell preserves its irreducibility). The four-cell variant additionally satisfies:

(CC1) Cooperative closure. For every cooperative capability $z \in P$, all participants of z lie in the same cell.

(CC2) Subsumption closure. No subsumption edge $c \succeq c'$ crosses a cell boundary: if c, c' are connected by subsumption in P , they lie in the same cell.

Under the cooperative-closure semantics for cell vol_P values—where each cell X is read as $X^* := X \cup \{z : z \text{ cooperative with all participants in } X\}$ and $\text{vol}_P(X)$ is summed over X^* —set-disjointness alone does not suffice for vol_P -additivity across cells. By the category-partition poset-independence condition of the companion paper [Lasser, 2026d, Revealed Sacrifice] and the M4 axiom of [Lasser, 2026f, Proposition 1], additivity requires full poset-independence—neither cross-cell cooperative composition nor cross-cell subsumption. A cooperative whose participants straddle two cells contributes to $\text{vol}_P(P)$ but to neither cell’s closure individually (CC1 failure); a subsumption edge crossing the boundary couples vol_P -mass across cells in ways M4’s independent-component additivity cannot decompose (CC2 failure). Together (CC1) and (CC2) give the poset-independence that licenses $\text{vol}_P(P) = \sum_X \text{vol}_P(X)$ for $X \in \{R, C, D, S\}$ under the cooperative-closure semantics. Strategy (c) below uses a literal-partition semantics instead, where each capability (cooperatives included) is assigned to exactly one cell by stipulation and cell-level vol_P values are read as atomic-weight accounting aggregates $\text{vol}_P(X) := \sum_{c \in X} w(c)$ rather than as the M1–M6 poset measure on poset-independent components; under that weaker semantics, the identity follows from direct rearrangement of the atomic sum across set-disjoint cells without the (CC1)/(CC2) hypotheses, but the cells are no longer M4-additive in the cooperative-closure sense. See Remark 13 for both regimes.

The δ terms are:

$$\begin{aligned}\delta_{\text{restricted}} &= [\text{vol}_P(R) - \text{vol}_R^{\text{lower}}(R)] / \text{vol}_P(P), \\ \delta_{\text{covered}} &= [\text{vol}_P(C) - \text{vol}_R^{\text{lower}}(C)] / \text{vol}_P(P), \\ \delta_{\text{dormant}} &= \text{vol}_P(D) / \text{vol}_P(P), \\ \delta_{\text{residual}} &= \text{vol}_P(S) / \text{vol}_P(P), \\ \delta_{\text{boundary}} &= [\text{vol}_P(B) - \text{vol}_R^{\text{lower}}(B)] / \text{vol}_P(P) \quad (\text{default strategy (c) only;} \\ &\quad \text{omit under (a)/(b)}).\end{aligned}$$

Each $\delta \geq 0$: $\text{vol}_R^{\text{lower}}(X) \leq \text{vol}_R(X) \leq \text{vol}_P(X)$ holds for any cell X (the first inequality by construction of $\text{vol}_R^{\text{lower}}$ as a lower bound on vol_R [Lasser, 2026d, Revealed Sacrifice], the second because vol_R is vol_P restricted to the exercised sub-poset and so cannot exceed vol_P on any subset), so the $\delta_{\text{restricted}}$ and δ_{covered} numerators are non-negative. The δ_{dormant} and δ_{residual} terms are non-negative by M5 ($\text{vol}_P \geq 0$). Two additivity stories underlie the identity, depending on closure strategy: (a)/(b) use M4 on poset-boundary-closed cells to obtain vol_P additivity, while (c) uses literal weight summation on a five-bucket set partition. These are not incompatible: (c) is a strictly weaker license that delivers the identity by direct summation without invoking M4 on cells, while (a)/(b) deliver the same identity via M4 on the four poset-boundary-closed cells. Under strategies (a) or (b), the identity $1 - \beta^{\text{lower}} = \sum \delta$ follows from two additivity steps. First, $\text{vol}_P(P) = \text{vol}_P(R) + \text{vol}_P(C) + \text{vol}_P(D) + \text{vol}_P(S)$ by poset-boundary closure plus set-disjoint union via M4 [Lasser, 2026f, Proposition 1]: CC1 prevents any cooperative from spanning two cells and CC2 prevents any subsumption from crossing, so the cells form an M4-additive partition for vol_P . Second, $\text{vol}_R^{\text{lower}}(P) = \text{vol}_R^{\text{lower}}(R) + \text{vol}_R^{\text{lower}}(C) + 0 + 0$ by the Part-B subset-extension of $\text{vol}_R^{\text{lower}}$, which under Part-B’s per-capability max-attribution [Lasser, 2026d, Revealed Sacrifice] is defined by $\text{vol}_R^{\text{lower}}(A) = \sum_{c \in A} \text{vol}_R^{\text{lower}}(c)$, making set-disjointness alone sufficient for additivity across cells; D and S contribute zero by definition (no admissible sacrifice evidence). CC1 and CC2 are not required for this $\text{vol}_R^{\text{lower}}$ step; they are required only for the vol_P step above. Under the literal-partition variant (strategy (c)), the vol_P identity expands to a five-term sum $\text{vol}_P(P) = \text{vol}_P(R) + \text{vol}_P(C) + \text{vol}_P(D) + \text{vol}_P(S) + \text{vol}_P(B)$ by literal atomic-weight

summation $\text{vol}_P(X) := \sum_{c \in X} w(c)$ across the five-cell partition; the $\text{vol}_R^{\text{lower}}$ aggregate similarly sums literally over the partitioned subset. $\text{vol}_R^{\text{lower}}(B)$ is computed by the same Part-B max-attribution rule as any other cell (it is not assumed zero) and equals zero only when no admissible bundle covers a boundary cooperative in B . This is the typical case (cross-cell cooperatives whose participants span at least one dormant or residual cell inherit zero $\text{vol}_R^{\text{lower}}$ contribution from those cells' absence of admissible bundle coverage), but it is a typical-case observation, not a definitional one; a cross-cell cooperative whose participants happen to be covered by an admissible bundle that contains the cooperative as a whole would contribute non-zero $\text{vol}_R^{\text{lower}}(B)$, and the δ_{boundary} formula $[\text{vol}_P(B) - \text{vol}_R^{\text{lower}}(B)]/\text{vol}_P(P)$ accommodates this case. Combining either route: $1 - \beta^{\text{lower}} = 1 - \text{vol}_R^{\text{lower}}(P)/\text{vol}_P(P) = [\text{vol}_P(P) - \text{vol}_R^{\text{lower}}(P)]/\text{vol}_P(P) = \sum_X [\text{vol}_P(X) - \text{vol}_R^{\text{lower}}(X)]/\text{vol}_P(P)$, which expands to the four-term identity under (a)/(b) (with $\text{vol}_R^{\text{lower}}(D) = \text{vol}_R^{\text{lower}}(S) = 0$ by the dormant/residual definitions, $B = \emptyset$) or the five-term identity under (c) (with $\text{vol}_R^{\text{lower}}(D) = \text{vol}_R^{\text{lower}}(S) = 0$ and $\text{vol}_R^{\text{lower}}(B)$ given by the max-attribution rule, typically zero).

Remark 12 (When the residual cell preserves its irreducibility under closure). *Under the default literal-partition strategy (c) the residual cell S always equals the seed $S_0 = R_S \setminus \Xi$ because (c) does not redistribute seed-classified atoms across cells (boundary-crossing cooperative nodes go to B , not to a participant's cell). The remainder of this remark addresses the cooperative-closure strategies (a)/(b), where seed-cell preservation is conditional. Under (a)/(b), the closure-adjusted residual cell S equals the seed S_0 when R_S is itself poset-isolated: no cooperative or subsumption edge connects a residual capability to a non-residual one. Under priority-extension, an edge from $c \in R_S \setminus \Xi$ to a non-residual node c' in C_0 , D_0 , or Ξ would absorb the entire connected component into the higher-priority cell of c' (residual having lowest priority); the absorbed mass then registers as δ_{covered} , δ_{dormant} , or $\delta_{\text{restricted}}$ rather than as δ_{residual} . The “residual is structurally irreducible” interpretation of Definition 12 therefore holds for the S -cell of the decomposition only when R_S is poset-isolated; when it is not, residual mass migrates to neighbouring cells under priority-extension. Operationally, R_S being poset-isolated is the typical case because the residual class characterizes purely-private exercise (Definition 12), which by construction does not participate in cooperative or subsumption structure with publicly-observable capabilities; verifying isolation is nonetheless an explicit residual-class hygiene check. Strategies (b) and (c) of Remark 13 preserve residual mass by other means: (b) sub-partitions cells by connected component, so residual components are accounted for separately; (c) sends boundary-crossing mass to a fifth bucket rather than absorbing it into a neighbouring cell.*

Remark 13 (Achieving poset-boundary closure of the cells). *Three operational strategies admit the poset-boundary closure hypothesis without changing the framework's substantive content. Each strategy handles both the cooperative (CC1) and subsumption (CC2) conditions jointly, since both are instances of the “edge connects different cells” pattern; concretely, treat the poset's combined cooperative/subsumption graph (each cooperative contributes hyperedges to its participants; each subsumption relation contributes a directed edge) as the object whose boundary-crossing instances must be eliminated.*

- (a) Priority-extension. For each node $n \in P$ not yet in a settled cell, and for each subsumption edge $c \succeq c'$ or cooperative z with participant set $\Pi(z)$ touching n : take the transitive closure under cooperative participation and subsumption edges, collect the resulting component, and place the whole component in the cell of its highest-priority member (restricted > covered > dormant > residual). After one sweep, the cells are both cooperative-closed and subsumption-closed (hence poset-boundary closed), and the original individual-capability classification is preserved except where a priority-dominated node was absorbed into a higher-priority cell. Caveat: dormant absorption. Priority-extension can absorb dormant or residual atomic capabilities into the covered cell when they are poset-connected to a covered atom. Their vol_P weight then contributes to $\text{vol}_P(C)$ but not to $\text{vol}_R^{\text{lower}}(C)$ (since they had no admissible sacrifice evidence by hypothesis), inflating δ_{covered} relative to a literal classification. This means δ_{covered} under priority-extension does not exclusively measure proxy-failure on observed capabilities; some of its mass may be absorbed dormancy or absorbed residual. Strategies (b) and (c) do not have this absorption effect, at the cost of weaker M4 license.

- (b) *Poset-closed sub-partition.* Refine the partition so each cell is sub-partitioned into its poset-connected components under both cooperative composition and subsumption globally across the entire P , not within the original cell only: the decomposition operates over the global connected-component partition of P , then assigns each component to a cell by its highest-priority member (as in strategy (a)). Sub-partitioning within an original cell alone does not eliminate edges crossing other original cells; the global-component view is what licenses M4. In practice strategy (b) coincides with strategy (a) on the global-component partition; the difference is presentational (sub-partitioning makes the component structure explicit before priority assignment).
- (c) *Boundary residual (literal-partition variant; default).* Define a literal set-partition of P into five buckets $\{R, C, D, S, B\}$ by stipulation: classify each capability into the highest-priority cell among its membership-determined seeds (as in (a) without the priority-extension absorption step), and place every boundary-crossing cooperative node (cooperatives in P whose participants span two or more original cells) into B rather than absorbing it into a participant's cell. Cooperative nodes carry explicit vol_P weight $w(c_{\text{coop}})$ from M5; the literal-partition assignment makes $\text{vol}_P(B) = \sum_{z \in \text{cross-cell coops}} w(z)$ by the strategy-(c) atomic-weight accounting convention. Subsumption-crossing edges, which carry no standalone vol_P weight, do not contribute additional bucket mass; their structural effect on additivity is what the literal-partition framing sidesteps. The five-cell identity

$$\text{vol}_P(P) = \text{vol}_P(R) + \text{vol}_P(C) + \text{vol}_P(D) + \text{vol}_P(S) + \text{vol}_P(B)$$

follows from literal weight summation over the partitioned capability census, without invoking M4-style poset-independence between cells. Under strategy (c) we adopt the atomic-weight accounting convention $\text{vol}_P(X) := \sum_{c \in X} w(c)$ for each cell X as an operational definition of cell-level vol_P values, distinct from vol_P 's general M1–M6 poset-measure semantics on poset-independent capability subsets. Cells partition P set-disjointly, so $\text{vol}_P(P) = \sum_X \text{vol}_P(X)$ follows by direct rearrangement of the atomic sum. This is a strictly weaker license than (a)'s M4-licensed identity: strategy (c) does not claim cross-cell poset-independence and does not certify M6-superadditivity properties of the partition; the cell-level vol_P values it produces are accounting aggregates, sufficient for the δ -decomposition but not interchangeable with the M1–M6 measure on poset-independent components. What it provides is the additive identity needed for the δ -decomposition, plus the diagnostic interpretation of δ_{boundary} as “cross-cell cooperative mass.” The identity becomes $1 - \beta^{\text{lower}} = \delta_{\text{restricted}} + \delta_{\text{covered}} + \delta_{\text{dormant}} + \delta_{\text{residual}} + \delta_{\text{boundary}}$ with all five terms well-defined as vol_P -fractions. Strategy (c) is the paper's default and the variant the worked example uses (Section 6): it preserves seed-cell attribution (dormant atoms stay in D rather than being absorbed into C as under (a)) so that δ_{covered} remains diagnostically interpretable as proxy-failure on observed capabilities and $\delta_{\text{dormant}}, \delta_{\text{residual}}$ retain their literal seed-classification meanings, at the cost of foregoing the stronger M4-additivity guarantee that (a) provides.

The paper's stated results assume strategy (c) by default: the literal-partition rule preserves the diagnostic-interpretation discipline that the worked example demonstrates, at the cost of a weaker (atomic-weight accounting only) additivity license. Strategies (a) and (b) remain available for deployments that prefer the stronger M4-additivity guarantee and accept the δ_{covered} absorption ambiguity in exchange; the practical prevalence of boundary-crossing poset edges is small when the capability partition's priority classes are themselves well-structured (e.g., restrictions are poset-boundary closed because restricting a cooperative implies restricting its participants, and restricting a capability implies restricting capabilities subsumed by it in cost-aligned posets), so the choice between (a)/(b) and (c) is largely a matter of which interpretive discipline the deployment requires.

Proposition 4 (Computability of Gap Decomposition, given precomputed governance inputs). *The complexity bound below is for the cell-classification step alone, given precomputed governance inputs as listed. Producing those inputs (admissibility labels via institutional attestation, residual classification by domain-expert review, governance-set restriction set Ξ and sufficiency thresholds) is itself a non-trivial pipeline whose cost is not part of this complexity claim. Given (i) the aggregate sacrifice data (events in $[T - H, T]$), (ii) the restriction set Ξ , (iii) the residual class R_S (Definition 12), (iv) the capability census (enumeration of P) together with the poset structure $\mathcal{M}$: the cooperative-composition hypergraph (cooperative nodes and their participants) and the subsumption relation graph (directed edges $c \succeq c'$), (v) an event-admissibility labelling designating*

which observed events satisfy the theorem-admissibility requirements. Because Definition 8 operates under Part B (the aggregate lower bound theorem [Lasser, 2026d, Revealed Sacrifice]), the full Part B admissibility set is required: $S0, S1, S2, S4(a), S3, S5$, the genuine-exchange conditions (i)–(iv) of the revealed-sacrifice event definition [Lasser, 2026d, Revealed Sacrifice], and within-bundle poset-independence per the event-local bundle decomposition [Lasser, 2026d, Revealed Sacrifice]; and (vi) a closure policy from Remark 13, with strategy (c) (literal-partition with boundary-residual bucket) as the default and (a)/(b) available as alternatives, the gap decomposition (five-term under (c), four-term under (a)/(b)) is computable in time $O(|P| + E + M \cdot \alpha(M))$, where E is the number of observed (capability, event) membership records, $M = |\mathcal{M}|$ is the total size of the poset structure (cooperative hyperedges plus subsumption edges), and $\alpha(\cdot)$ is the inverse Ackermann function from the amortized analysis of union-find. Since $\alpha(M)$ is ≤ 5 for any M realizable in physical computation, this is near-linear $\tilde{O}(|P| + E + M)$ for practical purposes. The $O(|P|)$ term is per-individual-capability classification; the $O(E)$ term is one-pass aggregation over bundle-incidence records; the $O(M \cdot \alpha(M))$ term is the closure-policy pass, which traverses cooperative and subsumption edges jointly via amortized union-find. No structural counterfactual computation is required (a direct-measurement approach would require $O(|P|^2)$ counterfactual queries).

Proof. *Pass 1 (individual-capability classification, $O(|P|)$).* Maintain a hash table from capabilities to their current class labels, initialized in $O(|P|)$ time. Sweep the individual capabilities in P once: for each $c \in P$, check $c \in \Xi$ (restricted table lookup, $O(1)$); if not, check $c \in R_S$ (residual classification lookup, $O(1)$); if in neither, mark as dormant. This gives an initial classification in $O(|P|)$.

Pass 2 (event-sweep, $O(E)$). Iterate through each (capability, event) incidence record. For each record (c, n) with event n admissibility-labelled satisfying the full Part B admissibility set of input (v)— $S0, S1, S2, S4(a), S3, S5$, the genuine-exchange conditions, and within-bundle poset-independence of Y_n per the event-local bundle decomposition [Lasser, 2026d, Revealed Sacrifice]—accumulate the event’s $\text{vol}_R^{\text{lower}}$ contribution to c regardless of c ’s priority class (so that $\text{vol}_R^{\text{lower}}(R)$ for restricted capabilities and $\text{vol}_R^{\text{lower}}(C)$ for covered capabilities are both maintained correctly per Definition 8’s δ formulas, which subtract $\text{vol}_R^{\text{lower}}(R)$ and $\text{vol}_R^{\text{lower}}(C)$ respectively). The class label set in Pass 1 is preserved (restricted $\rightarrow$ restricted, residual $\rightarrow$ residual, dormant $\rightarrow$ covered when receiving evidence); only the $\text{vol}_R^{\text{lower}}$ accumulation runs unconditionally. Events failing any component of the Part B admissibility set do not contribute coverage; in particular, events with internally cooperative bundles are filtered out by the within-bundle poset-independence check. This is $O(E)$ total work, one constant-time update per record.

Pass 3 (closure-policy pass, $O(M)$). Under default strategy (c), walk the cooperative hyperedges identifying boundary-crossing cooperative nodes (cooperatives whose participants span two or more original cells) and place each into the boundary-residual bucket B ; non-boundary-crossing cooperatives remain in their participants’ cell. Subsumption edges contribute no vol_P weight and are therefore not processed; the literal-partition framing sidesteps their additivity contribution. Under alternative strategy (a), walk the combined cooperative/subsumption graph via union-find, identifying poset-connected components: two nodes are in the same component if connected by any chain of cooperative-participation or subsumption edges. For each component, determine its highest-priority cell (restricted $>$ covered $>$ dormant $>$ residual) among member cells, and place the entire component in that cell. The cells become both cooperative-closed and subsumption-closed (hence poset-boundary closed per Definition 8) by construction. Union-find gives amortized $O(M \cdot \alpha(M))$ complexity, essentially linear in M . Under strategy (b), the pass is replaced by the sub-partition construction over the same combined graph, with the same complexity modulo constants. Strategy (c)’s single hyperedge sweep is $O(M)$ outright (no union-find needed since no transitive-closure of cell membership is performed).

Aggregation. Each class’s aggregate vol_P and $\text{vol}_R^{\text{lower}}$ is maintained incrementally across the three passes; computing the four quotients from the running totals at the end of Pass 3 is $O(|\text{cells}|) = O(1)$ since there is a constant number of cells, but the running totals themselves are updated as nodes move between cells under closure (Pass 3), so the total accounting cost is folded into Passes 1–3 rather than literally appended. The counterfactual comparison (removing capability c and recomputing global reachability) is not performed; only membership checks, aggregation, and closure-policy lookups occur. A direct-measurement approach requiring $O(|P|^2)$ counterfactuals is avoided because cov-

ered/dormant classification here uses bundle-appearance evidence, not marginal-contribution evidence. ■

Remark 14 (Admissibility-labelling is an external input). *S0 (calibration) and the interior component of S1 are not mechanically verifiable from the observable event fields alone. Input (v) of Proposition 4 represents an external admissibility layer: events must be labelled admissible (by trust-weighted attestation, auditor sampling, or explicit agent commitment) before the decomposition can classify capabilities into “covered.” Under committed-mode operation ([Lasser, 2026d, Revealed Sacrifice]), the labelling is carried by the commitment protocol’s proof attachment; in unrestricted mode, the framework must rely on governance processes to supply it. The $O(|P|)$ complexity is the classification cost given the labelling, not the cost of producing the labelling itself.*

Remark 15 (Part A versus Part B granularity). *Under Part A alone (category-level aggregates, requiring S0, S1, S2, S4(a), and the genuine-exchange conditions of the revealed-sacrifice event definition in the companion paper [Lasser, 2026d, Revealed Sacrifice]), the decomposition operates at the category level: each partition component C_j (the category partition [Lasser, 2026d, Revealed Sacrifice]) is classified as restricted, covered, dormant, or residual. Under committed data ([Lasser, 2026d, Revealed Sacrifice]), S2 splits into two components with different attestation routes: (i) the objective side, asserting that $\Delta\text{vol}_P(X)$ is correctly computed from the surrendered benchmarked capability via the public benchmark function, is verified by the commitment layer’s ZK proof on the magnitude commitment; (ii) the subjective side, asserting that the agent’s valuation $U_i(X)$ of the surrendered capability matches $\Delta\text{vol}_P(X)$ in the calibration scale, is not ZK-verifiable and must be supplied by external attestation (see the external-attestation remark in the companion paper [Lasser, 2026d, Revealed Sacrifice]). The committed-mode decomposition therefore certifies the objective S2 component via the commitment layer and assumes the subjective S2 component via the attestation layer. The per-capability decomposition is the refinement available when Part B assumptions hold.*

3.3 Need-sufficiency diagnostic (companion metric)

The gap decomposition above tracks want-pursuit: how much of the collective’s above-sufficiency capability-space is evidenced by voluntary sacrifice. The need-sufficiency diagnostic is the parallel metric for below-sufficiency: how efficiently does the collective satisfy needs, and is the satisfaction downstream-safe?

Definition 9 (Need-Access Cost). *The need-access cost aggregate for trade window $[T - H, T]$ is:*

$$\text{NAC}(T) = \sum_{n \in F_{S1}} \Delta\text{vol}_P(X_n),$$

where F_{S1} is the set of events failing S1 due to below-sufficiency (Definition 6). NAC measures the total benchmarked capability surrendered for need-satisfaction—purchasing power and time spent reaching baseline functionality rather than pursuing wants.

NAC does not enter the aggregate lower bound of the companion paper [Lasser, 2026d, Revealed Sacrifice] or β^{lower} . It is a separate diagnostic that tracks the cost side of the agent’s capability budget.

Remark 16 (NAC is an upper bound on need-access cost). *Because the whole-event filter sends the entire $\Delta\text{vol}_P(X_n)$ of mixed need/want events into NAC, NAC overestimates the true need-access cost by including any attached want-pursuit spend. NAC should be read as an upper bound on need-access cost, not a precise measurement.*

Looseness magnitude depends on the deployment’s mixed-event prevalence. In economies where most below-sufficiency events involve pure-need bundles (a clinic visit purely for medical care, a bottle of water purely for hydration), NAC is close to the true need-access cost. In economies where below-sufficiency events typically carry attached want-pursuit (a grocery basket that contains both staple food at below-sufficiency-cost wages and discretionary items, a transit pass that provides both need-satisfaction and recreational mobility), NAC can substantially overstate the true need-access cost, potentially by factors of two or more in dense mixed-bundle markets. The diagnostic interpretation of “high NAC alarms” in Proposition 5’s quadrants is therefore sensitive to the deployment’s mixed-bundle prevalence: a low-mixed-prevalence deployment can read NAC thresholds nearly literally; a high-mixed-prevalence deployment should treat NAC thresholds as conservative upper-bound triggers that may fire on inflated values rather than on true need-access cost increases.

A bundle-decomposition extension that splits $\Delta \text{vol}_P(X_n)$ into need-satisfying and want-pursuing components via the event-local bundle decomposition of the companion paper [Lasser, 2026d, Revealed Sacrifice] would tighten NAC to the need-component aggregate and is the natural follow-up; it is not part of this paper’s NAC definition because the event-local decomposition coefficients $\alpha_{n,c}$ are not generally available at the granularity required without additional governance attestation.

Definition 10 (Need-Satisfaction Efficiency). *For each need N identified in the poset (Definition 1) and each agent i , the per-agent need-satisfaction efficiency is a triple:*

$$\text{NSE}(i, N) = (\text{sufficiency_status}, \text{downstream_safety}, \text{access_cost}),$$

where:

- $\text{sufficiency_status} \in \{\text{below}, \text{met}, \text{exceeded}\};$
- $\text{downstream_safety} \in \{\text{safe}, \text{degrading}, \text{unknown}\};$
- $\text{access_cost} = \sum \Delta \text{vol}_P(X_n)$ for below-sufficiency events by agent i targeting need N .

The collective-level aggregate $\text{NSE}(T)$ reports the distribution of (agent, need) pairs across sufficiency statuses and flags degrading pathways.

Remark 17 (Observability under commitment). *When sacrifice events are committed ([Lasser, 2026d, Revealed Sacrifice]), per-agent $\text{NSE}(i, N)$ is not recoverable. The committed-mode fallback is a population-level proxy: $\text{NSE}_{\text{proxy}}(T)$ reports $\text{NAC}(T)$ by need category and counts of below-sufficiency events. Full $\text{NSE}(T)$ requires agents to voluntarily attest pathways or the framework to operate in non-committed mode.*

Proposition 5 (Combined Diagnostic). *$\text{vol}_R^{\text{lower}}$ and NAC are not directly commensurable as written: $\text{vol}_R^{\text{lower}}$ is a poset-measure quantity bounded above by $\text{vol}_P(P)$, while NAC is a cumulative event aggregate of $\Delta \text{vol}_P(X_n)$ summands that can grow without bound as more below-sufficiency events accumulate within the window. For diagnostic comparison, we use the normalised pair $(\text{vol}_R^{\text{lower}}/\text{vol}_P(P), \text{NAC}/B_W)$ where B_W is a domain-supplied total benchmarked-budget scale for window W (e.g., aggregate purchasing power exchanged in W , or a normalised-per-agent equivalent). “High” and “low” below refer to these normalised quantities crossing governance-supplied thresholds, not to the raw aggregates. The framework reports the triple $(\text{vol}_R^{\text{lower}}, \text{NAC}, \text{NSE})$ as a joint diagnostic:*

- (a) **High $\text{vol}_R^{\text{lower}}$ + low NAC:** *Healthy. The collective exercises rich capabilities above sufficiency and needs are cheap to meet.*
- (b) **Low $\text{vol}_R^{\text{lower}}$ + high NAC:** *Proxy failure candidate. vol_P may be high, but agents are spending most of their benchmarked capability budget on basic need-satisfaction. The sacrifice channel excludes the H1+H2 below-sufficiency sub-class characterized by Definition 6 (retention degrading and no third non-degrading alternative, i.e., the sub-class where S1 unambiguously fails), so a low $\text{vol}_R^{\text{lower}}$ alongside a high NAC is consistent with low above-sufficiency exercise; the gap-decomposition’s alarm-interpretation caveats (Remark 11) still apply (the low $\text{vol}_R^{\text{lower}}$ may also reflect bound-looseness, sparse observation, or admissibility filtering). NAC surfaces a candidate explanation for the low $\text{vol}_R^{\text{lower}}$.*
- (c) **Low $\text{vol}_R^{\text{lower}}$ + low NAC:** *Dormancy, residual, or theorem-inadmissible activity. The gap decomposition (Definition 8) distinguishes dormant from residual.*
- (d) **High $\text{vol}_R^{\text{lower}}$ + high NAC:** *Mixed regime. NSE distinguishes “expensive but safe” from “expensive and degrading.”*

Remark 18 (Parallel structure, not nesting). *The need-sufficiency diagnostic is parallel to the gap decomposition, not nested within it, but operates on different formal objects. The gap decomposition is a capability partition; NAC is an event aggregate; NSE is a per-need state assessment. The two diagnostics cover complementary portions of the sacrifice event stream: S1-satisfying events (that also satisfy S0 and S2) enter the gap decomposition via the aggregate lower bound theorem [Lasser, 2026d, Revealed Sacrifice]; events failing S1 due to below-sufficiency enter NAC. Events failing S1 for non-need reasons (coercion, duress) are excluded from both diagnostics and constitute an undiagnosed event-stream residual.*

4 Wireheading-Consistent Signature via Trade-Flow Concentration

Wireheading produces high β^{lower} with degenerate exercise: the collective exercises capabilities, but only in narrow self-reinforcing loops. Under the sacrifice-based observation model, the wireheading signature appears as *trade-flow concentration*: the collective’s sacrifice events cluster in a narrow category of unbenchmarked bundles, rather than distributing across the capability space.

Definition 11 (Trade-Flow Leverage). *For bundle category $C_j \in \mathcal{C}$ (the category partition [Lasser, 2026d, Revealed Sacrifice]), the trade-flow leverage of C_j at time T is:*

$$\lambda_{\text{trade}}(j, T) = \begin{cases} \text{vol}_R^{\text{lower}}(C_j, T) / \text{vol}_R^{\text{lower}}(P, T) & \text{if } \text{vol}_R^{\text{lower}}(P, T) > 0, \\ \text{undefined (diagnostic suppressed)} & \text{if } \text{vol}_R^{\text{lower}}(P, T) = 0. \end{cases}$$

The category-level subset aggregate $\text{vol}_R^{\text{lower}}(C_j, T)$ is theorem-certified as a Part-B lower bound only when $C_j \cap U$ is internally poset-independent ([Lasser, 2026d, Revealed Sacrifice], subset-extension proposition); the cross-category poset-independence between distinct C_j, C_k given by the category partition’s poset-independence condition [Lasser, 2026d, Revealed Sacrifice] is a separate condition (across categories, not within them) and does not by itself certify within-category aggregation. Where within-category poset-independence does not hold, $\text{vol}_R^{\text{lower}}(C_j, T)$ should be read as an accounting score over the per-capability max-attribution sum on $C_j \cap U$ rather than a theorem-certified Part-B aggregate; the leverage ratio λ_{trade} remains diagnostically meaningful in that regime as a relative concentration signal but its denominator interpretation weakens accordingly. Under healthy trade diversity, leverage is distributed across many categories. Under wireheading, leverage concentrates on the loop drivers. When total observed trade flow is zero (the agent has produced no theorem-admissible sacrifice events in the window), λ_{trade} is not a meaningful quantity and the diagnostic is suppressed; the relevant signal in that regime is the gap decomposition (§3), which classifies capabilities as dormant or residual rather than as concentration-anomalies.

Proposition 6 (Wireheading-Consistent Concentration Signal). *When $\text{vol}_R^{\text{lower}}(P, T) > 0$, the Herfindahl–Hirschman Index [Rhoades, 1993]*

$$\text{HHI}(\lambda_{\text{trade}}) = \sum_j \lambda_{\text{trade}}(j)^2$$

is Schur-convex on the simplex of probability vectors: for leverage distributions λ, λ' over the same support, if λ majorizes λ' (i.e., λ is at least as concentrated as λ' in the sense of Hardy et al. [1929]), then $\text{HHI}(\lambda) \geq \text{HHI}(\lambda')$. Concentration in the majorization order therefore induces a monotone ordering of HHI values, and comparing HHI against a governance-set threshold $\text{HHI}_{\text{alarm}}$ provides a concentration-consistent-with-wireheading alarm: if $\text{HHI} > \text{HHI}_{\text{alarm}}$, a small number of bundle categories dominate the aggregate, a pattern consistent with (but not exclusive to) narrow self-reinforcing exercise. The Schur-convexity result establishes monotonicity in the majorization order; the inference from concentration to wireheading requires additional context (governance threshold, domain baseline, cross-diagnostic confirmation as in Proposition 5).

Remark 19 ($\text{HHI}_{\text{alarm}}$ is a governance parameter, not a theorem-derived separator). *The alarm threshold $\text{HHI}_{\text{alarm}}$ is not derived from the framework’s axioms; it is a governance parameter calibrated to the framework’s tolerance for concentration risk, analogous to HHI thresholds in antitrust practice and to the sufficiency thresholds of Definition 1. Its appropriate value depends on the support size (number of categories with non-zero leverage), the domain’s expected baseline concentration, and the framework’s false-positive/false-negative tolerance. The structural guarantee is that HHI is monotone in concentration; the separation between wireheading-consistent and wireheading-inconsistent patterns is operational, not theorem-derived.*

Proof. Let $m = |\{j : \text{vol}_R^{\text{lower}}(C_j) > 0\}|$ be the number of bundle categories with non-zero observed leverage. The leverage vector $\lambda = (\lambda_{\text{trade}}(j))_j$ is a probability distribution: $\lambda_j \geq 0$ and $\sum_j \lambda_j = 1$ by the normalization in Definition 11.

Schur-convexity. $\text{HHI}(\lambda) = \sum_j \lambda_j^2$ is a sum of convex functions of the coordinates and is symmetric under permutation. Any symmetric convex function on the simplex is Schur-convex by the Hardy–Littlewood–Pólya theorem [Hardy et al., 1929, Marshall et al., 2011]; therefore λ majorizing λ'

implies $\text{HHI}(\lambda) \geq \text{HHI}(\lambda')$. Equivalently, spreading mass from heavier bins to lighter bins weakly decreases HHI, and concentrating mass from lighter bins to heavier bins weakly increases it.

Quantitative bounds. For any probability distribution λ with support of size m , Cauchy–Schwarz (or Jensen’s inequality on the squaring map) gives $\text{HHI}(\lambda) \geq 1/m$, with equality iff λ is uniform—the minimally-concentrated distribution in the majorization order. At the other extreme, if the top k categories capture fraction ρ of total leverage, $\text{HHI} \geq \rho^2/k$ from the top- k mass alone (concentrating that mass uniformly on the k heavy bins). When $k < m$ and $\rho < 1$, the full HHI also includes a tail contribution $\geq (1 - \rho)^2/(m - k)$ from the remaining mass spread across the $m - k$ tail bins (Cauchy–Schwarz applied to the tail), giving $\text{HHI} \geq \rho^2/k + (1 - \rho)^2/(m - k)$. When $k = m$ (top set is the whole support) the tail bound is vacuous and the bound reduces to $\text{HHI} \geq \rho^2/k = 1/m$, recovering the uniform lower bound; when $\rho = 1$ the top set absorbs all mass and again the tail term vanishes. The simpler ρ^2/k bound is what the alarm condition uses in practice; the tail term is small for highly concentrated distributions (ρ near 1 or $m \gg k$) and its inclusion only strengthens the lower bound.

Alarm criterion. A wireheading signature concentrates trade flow in k loop-driver categories with ρ close to 1 and k small, forcing $\text{HHI} \geq \rho^2/k$. A governance-chosen threshold $\text{HHI}_{\text{alarm}} \in [1/m, 1)$ when $m \geq 2$, and the alarm degenerates to “always fires” when $m = 1$ (a single active category with $\text{HHI} = 1$ exceeds any threshold below 1); the latter case is the limiting wireheading signature where every observed event is in the same category, so the auto-alarm matches the diagnostic intent. The threshold determines which concentration levels trigger the alarm; by Schur-convexity, the alarm fires on a majorization-upper set of distributions (if λ triggers the alarm and $\lambda' \succeq \lambda$ in the majorization order, then λ' also triggers it). Whether any specific $\text{HHI}_{\text{alarm}}$ cleanly separates “diverse” from “loop-dominated” depends on the domain’s baseline concentration and the framework’s tolerance for false positives/negatives—the threshold is a governance parameter, not a theorem-derived separator (see Remark 19). ■

Remark 20 (Advantage over structural counterfactual). *A direct-measurement approach would try to identify wireheading via exercise-leverage concentration, requiring the marginal vol_R contribution of each capability (a counterfactual query). The trade-flow formulation surfaces the concentration signal via HHI on observed trade data, requiring no counterfactual computation. The diagnostic is strictly more tractable; it remains a signal rather than a definitive detection, per Remark 19.*

Proposition 7 (Third-Party Observability of the Concentration Signal (Part A)). *Restrict attention to events that have been labelled S1-admissible by the institutional attestation layer (the ones that pass below-sufficiency filtering and the other S1 hypotheses; the unfiltered public ledger may contain events that fail S1, and including those would let need-driven cost-of-access spending drive the concentration statistic, conflating wireheading with poverty). When the total S1-admissible sacrifice magnitude $\sum_{n: n \text{ S1-admissible}} \Delta \text{vol}_P(X_n) > 0$, the Part A category-flow concentration statistic $\widehat{\text{HHI}}$ is computable from public trade data (committed sacrifice events, the committed event definition [Lasser, 2026d, Revealed Sacrifice]) restricted to S1-admissible events, without access to any agent’s private state beyond the admissibility label itself (which is a public attestation, not interior state). Without the S1-admissibility filter, HHI remains computable but loses its wireheading-consistent reading: a high HHI on the unfiltered ledger could reflect concentrated need-driven below-sufficiency sacrifice rather than narrow want-pursuit loops. The Part A statistic is built on the aggregate sacrifice mass $\Delta \text{vol}_P(X_n)$, not on the Part B per-capability leverage $\text{vol}_R^{\text{lower}}(C_j, T)$ of Definition 11; the two coincide only when the per-event Part A bound $\text{vol}_R(Y_n) \geq \Delta \text{vol}_P(X_n)$ is exactly proportional to the per-category Part B aggregate, which is not generally the case (Part B requires within-bundle poset-independence and the additional decomposition assumptions S3 and S5). $\widehat{\text{HHI}}$ is therefore a sacrifice-mass concentration statistic that admits the wireheading-consistent reading of Proposition 6 via the same Schur-convexity argument applied to a different normalised distribution; it is not a public computation of the Part B HHI. Let $\mathcal{E}_{S1}$ denote the set of S1-admissible events in the trade window (events that have passed the institutional admissibility-attestation pipeline). Specifically, define*

$$\tilde{\lambda}_{\text{trade}}(j, T) = \begin{cases} \frac{\sum_{n \in \mathcal{E}_{S1}: \text{category}(Y_n)=j} \Delta \text{vol}_P(X_n)}{\sum_{n \in \mathcal{E}_{S1}} \Delta \text{vol}_P(X_n)} & \text{if } \sum_{n \in \mathcal{E}_{S1}} \Delta \text{vol}_P(X_n) > 0, \\ \text{undefined (diagnostic suppressed)} & \text{if } \sum_{n \in \mathcal{E}_{S1}} \Delta \text{vol}_P(X_n) = 0, \end{cases}$$

and

$$\widetilde{\text{HHI}} = \sum_j \tilde{\lambda}_{\text{trade}}(j)^2,$$

for the trade window T (with $\widetilde{\text{HHI}}$ correspondingly suppressed when $\tilde{\lambda}_{\text{trade}}$ is undefined). A third-party auditor with access only to the committed sacrifice ledger plus the admissibility-label attestations (which are public attestation outputs, not the agent’s interior state) can compute $\widetilde{\text{HHI}}$ whenever the window contains positive-magnitude S1-admissible events and raise the Part A category-flow alarm (wireheading-consistent in the sense of Proposition 6, with $\tilde{\lambda}_{\text{trade}}$ substituted for λ_{trade} ; not a wireheading determination on its own). “Third-party observable” here means the auditor can compute the statistic from public inputs; it does not mean the statistic is self-verifying—the admissibility labels remain external attestations whose correctness is the responsibility of the attestation layer (Remark of the companion paper’s external-attestation model in [Lasser, 2026d, Revealed Sacrifice]). When the window has zero total sacrifice flow, the diagnostic is suppressed (the relevant signal in that regime is the gap decomposition, §3, matching the parallel suppression rule of Definition 11).

Proof. By the committed event definition [Lasser, 2026d, Revealed Sacrifice], every committed sacrifice event exposes the pair $(\text{category}(Y_n), \Delta\text{vol}_P(X_n))$ in cleartext. For the category-flow statistic, both the numerator $\sum_{n \in \mathcal{E}_{S1}: \text{category}(Y_n)=j} \Delta\text{vol}_P(X_n)$ and the denominator $\sum_{n \in \mathcal{E}_{S1}} \Delta\text{vol}_P(X_n)$ are direct functions of the revealed-cleartext fields plus the public S1-admissibility attestation labels (which mark which committed events belong to $\mathcal{E}_{S1}$). No hidden field—agent identity, bundle-interior subset, $\alpha_{n,c}$ coefficients—enters the computation. Hence $\widetilde{\text{HHI}} = \sum_j \tilde{\lambda}_{\text{trade}}(j)^2$ is publicly computable from the committed-event ledger plus the admissibility attestation labels. The alarm test $\widetilde{\text{HHI}} > \text{HHI}_{\text{alarm}}$ is consequently computable by any party with read access to both the committed-event ledger and the admissibility attestations. Monotonicity of $\widetilde{\text{HHI}}$ in trade-flow concentration across categories follows from the same Cauchy–Schwarz / Herfindahl argument as Proposition 6, applied to the category-flow distribution. ■

Remark 21 (Part B observability requires richer commitments). *The per-capability Part B concentration statistic $\text{HHI}(\lambda_{\text{trade}})$ built on $\text{vol}_R^{\text{lower}}(C_j, T) = \sum_{c \in C_j \cap U} \text{vol}_R^{\text{lower}}(c, T)$ is not third-party computable from the committed-event interface of the committed event definition [Lasser, 2026d, Revealed Sacrifice]. The committed form reveals only the partition label $\text{category}(Y_n)$ and attests $Y_n \subseteq C_{\text{category}(Y_n)}$; it does not reveal which subset of $C_{\text{category}(Y_n)}$ was actually acquired, nor the per-capability attribution coefficients $\alpha_{n,c}$. Computing $\text{vol}_R^{\text{lower}}(c, T) = \max_{n: c \in Y_n} \alpha_{n,c} \cdot \Delta\text{vol}_P(X_n)$ therefore requires additional committed fields: per-event subset-membership openings (or a ZK attestation equivalent), and the $\alpha_{n,c}$ coefficients (public by an equal-weight default, or committed alongside the event under a richer Part B variant as flagged in the Part B commitment remark [Lasser, 2026d, Revealed Sacrifice]). Under that richer commitment, the Part B statistic becomes third-party observable from the public committed events plus the public S1-admissibility labelling; under the default Part A commitment, it is not. Within a consented operational perimeter O (the exercise-indicator appendix of [Lasser, 2026d, Revealed Sacrifice]), no richer commitment is needed because direct per-capability evaluation is permitted; the Part B statistic is then available to the perimeter operator, not to arbitrary third parties.*

5 Residual Class Under Revealed-Sacrifice Observation

5.1 Re-characterization

A natural alternative approach would define the residual class as capabilities failing benchmarkability (identity-individuated, non-repeatable, non-communicable). The revealed-sacrifice framing produces a differently-shaped residual class that is, by our interpretation, more naturally aligned with sacrifice-channel diagnostics: the channel-reach criterion is anchored in the observation infrastructure’s modality (what kinds of events the channel emits, expressed as a structural claim about candidate exercise pathways), while the benchmarkability criterion is anchored in capability-individuation discipline. Both criteria require domain analysis to apply to specific capabilities (see the structural/domain proof-obligation note in Definition 12 below); neither is *directly checkable*

from trade records alone. Whether the revealed-sacrifice framing constitutes “operationally cleaner” is an interpretation, not a theorem.

Definition 12 (Residual Class Under Revealed Sacrifice). *The residual class is the set of capabilities structurally outside the reach of the sacrifice channel—capabilities whose exercise, by their nature, produces no sacrifice event even with unlimited observation time:*

$$R_S = \{c \in P : \text{no sacrifice-observable path exists in principle}\}.$$

R_S is a structural property of the capability, not a property of the observation window. A capability is in R_S because the sacrifice channel cannot reach it (purely private exercise with no material footprint), not because the channel has not yet observed it. Establishing R_S membership is a structural/domain proof obligation, not a directly checkable property of trade infrastructure: the analyst must show that every conceivable exercise mode of the candidate capability fails to produce a sacrifice event, which involves enumeration over hypothetical exercise pathways rather than inspection of observed trades. Trade infrastructure shows what happened; R_S membership claims what cannot happen in principle. Examples are restricted to capabilities that admit no sacrifice-observable exercise mode in principle: private contemplative experience whose exercise consumes neither paid material nor time-displaced opportunity beyond idle cycles, and passively-concurrent internal capabilities such as ongoing interoceptive self-awareness exercised concurrently with all other activity (see proof of Proposition 8(c) for a worked witness). Cognitive capabilities that admit paid-training, dedicated-time, or counterparty-supported exercise modes (silent rehearsal as a paid course, planning sessions on deliberately blocked time, supervised reasoning practice) lie outside R_S because at least one sacrifice-observable mode exists in principle, even if no specific window observes it.

Remark 22 (Structural versus evidential absence). R_S is distinct from the set of capabilities with no sacrifice evidence in a given window. A capability outside R_S (tradable in principle) may have no sacrifice evidence in a particular window simply because no trade happened to cover it—this is the dormant category in the gap decomposition (Definition 8), not the residual class. Dormant capabilities can be resolved by waiting for more trade data; residual capabilities cannot.

Remark 23 (Conservatism of R_S classification). Classifying a capability as residual should be conservative. “Plausible mechanism” is itself contestable: “private contemplative experience” might displace idle cycles that have time-opportunity cost, “baseline” awareness capabilities might be measurable through clinical assessment that costs the agent’s time. Classify $c \in R_S$ only when every proposed sacrifice path can be shown to belong to a distinct capability (e.g., the act of being assessed is a distinct capability from the assessed underlying state) under the framework’s poset-construction discipline. This shrinks R_S relative to a looser reading; a smaller residual class makes the gap-decomposition’s residual cell smaller, which is preferable since δ_{residual} is the irreducible component beyond the channel’s reach. In practice, residual-class membership claims should be accompanied by an explicit argument that all candidate sacrifice paths attach to distinct capabilities, not the target c itself. When such arguments are absent, classify c as non-residual by default.

If there is any plausible mechanism by which exercise of c could produce a sacrifice event (even indirectly—e.g., an artistic practice that requires purchasing supplies), c should be classified as outside R_S . The residual class should be small by construction.

R_S is an analyst-dependent governance parameter, not a structural fact certifiable from trade infrastructure. The “in principle” criterion of Definition 12 requires a domain analyst to adjudicate which proposed sacrifice paths attach to the target capability and which to distinct capabilities; reasonable analysts will disagree on cases (e.g., does “baseline interoceptive self-awareness” admit a clinical-assessment sacrifice path that attaches to it, or does the assessment attach to a distinct “measured-awareness” capability under the individuation discipline?). Proposition 8’s conditional clause makes this explicit: whether $R_S \setminus R_B$ is non-empty depends on individuation choices. δ_{residual} should therefore be read as a governance-parametrized quantity whose magnitude encodes the deployment’s residual-class adjudication discipline, not as a structural fact independent of analyst choice. Two deployments operating on the same trade pool under different individuation conventions can produce different δ_{residual} ; the gap decomposition itself is robust to the choice (the additive identity holds under any consistent classification), but cross-deployment δ_{residual} comparisons require the classification conventions to be stated.

5.2 Relationship to the benchmarkability-based residual class

Proposition 8 (Residual Class Intersection). *Let R_B be the benchmarkability-based residual class (capabilities failing benchmarkability under a direct-measurement approach) and R_S be the revealed-sacrifice residual class (Definition 12). Then:*

- (a) $R_B \setminus R_S$ is non-empty (witnessed in (b) below).
- (b) Capabilities in $R_B \setminus R_S$: identity-individuated capabilities that nonetheless involve observable sacrifice. Example: an artistic practice that is identity-individuated but involves purchasing supplies, paying for studio space, foregoing labour hours. The time-sacrifice channel captures a lower bound even though the capability is identity-individuated.
- (c) Whether $R_S \setminus R_B$ is non-empty is individuation-dependent: under poset-construction choices that distinguish baseline from trained/honed forms of internal capabilities (treating them as distinct nodes), candidate witnesses such as baseline interoceptive self-awareness lie in $R_S \setminus R_B$; under individuation choices that treat baseline and trained as co-extensive, no witness in $R_S \setminus R_B$ has been established by this paper. See (c) of the proof.
- (d) Whether $R_B \cap R_S$ is non-empty is individuation-conditional in the same sense as (c): candidate witnesses (private contemplative experience with no material input or time-sacrifice signal) lie in $R_B \cap R_S$ under poset-construction choices that treat “idle” time as not opportunity-cost-attributable; under a stricter individuation that attributes opportunity cost even to idle intervals, the witness migrates out of R_S . See (d) of the proof for the conditional witness.

The proposition’s downstream uses in this paper require only (a) and (b); (c) and (d) are presented as individuation-conditional witnesses for completeness, and $R_B \cap R_S$ is reported as conditionally non-empty rather than unconditionally non-empty. The classical and channel-reach residual carves therefore overlap conditionally (witnessed by (d) under the stipulated individuation) but are not in a containment relationship in the strict direction $R_B \subseteq R_S$ (witnessed by (b)); the converse direction $R_S \subseteq R_B$ is not asserted by this paper.

Proof. Each part is established by exhibiting a capability with the claimed membership, which proves non-containment in the relevant direction.

(a) $R_B \setminus R_S$ non-empty (witnessed in (b)) ruling out $R_B \subseteq R_S$. Part (b) below produces an explicit witness for $R_B \setminus R_S \neq \emptyset$. The reverse direction $R_S \subseteq R_B$ is not established by this proposition (whether it holds depends on the conditional witness construction in (c) below); the proposition therefore asserts only the asymmetric statement that R_B is not a subset of R_S , plus the intersection-non-empty witness in (d).

(b) $R_B \setminus R_S$. Consider the capability “a specific artist’s studio practice.” Under the benchmarkability criterion of [Lasser, 2026f, Definition 2], this is identity-individuated: the artefact sequence is non-repeatable and non-communicable in the benchmark sense, so the capability lies in R_B . But the practice consumes materials (paid-for supplies, rent on studio space) and displaces outside-option labour hours; every session admits a money- or time-sacrifice event ([Lasser, 2026d, Revealed Sacrifice]) on the underlying bundle. Whether that event ultimately emits a $\text{vol}_R^{\text{lower}}$ signal additionally requires the sacrifice channel’s S0–S5 admissibility and genuine-exchange conditions to hold for the event; what R_S -membership turns on is the *reachability in principle* of a sacrifice-observable path, not the admissibility of any particular event. Reachability is satisfied here because each session *admits* (in the reachability-in-principle sense—not that S0–S5 are necessarily verified for each event) a sacrifice event under standard market conditions; therefore the capability is reachable in principle by the sacrifice channel and so $\notin R_S$.

(c) $R_S \setminus R_B$. The residual definition (Definition 12) requires that *no* exercise mode produces a sacrifice event in principle—not merely that some modes do not. Many cognitive capabilities that feel purely internal (silent rehearsal, planning, reasoning) admit paid-training or dedicated-time exercise modes and therefore lie outside R_S . The genuine witnesses for $R_S \setminus R_B$ are capabilities exercised passively and concurrently with other activity, with no incremental material input, no incremental time displacement, and no counterparty.

$R_S \setminus R_B$ witnesses are harder to construct than $R_B \setminus R_S$ because clinical measurement, training programs, and structured-report assessments typically generate sacrifice events on capabilities that *seem* purely internal, pulling such capabilities out of R_S .

Consider “*baseline* interoceptive self-awareness as a background condition of conscious embodiment, distinguished from *trained interoceptive acuity* which clinical neuroscience can measure.” Trained acuity is benchmarkable *and* reachable by the sacrifice channel through paid-training programs, mindfulness courses with explicit fees, and clinically-billable assessment sessions, so trained acuity lies in neither R_S nor R_B . The candidate $R_S \setminus R_B$ witness is the *baseline* component itself. The load-bearing claim is that *incrementally exercising the baseline* (as distinct from training a higher acuity level, or measuring the current acuity) requires no sacrifice—the baseline is on whenever the agent is conscious, and any sacrifice associated with clinical measurement or training accrues to the *measurement* or *training* capability, not to the baseline-exercise capability. This separation is contestable: a stricter reading would say baseline interoceptive exercise is co-extensive with trained interoceptive acuity exercise (any clinical measurement is also a measurement of the baseline, hence is a sacrifice path to the baseline), in which case the baseline is reachable in principle and lies outside R_S . Whether the separation holds is a domain question about how interoceptive capability is individuated. We adopt the separation: under the framework’s poset-construction discipline, an analyst declaring “baseline interoceptive self-awareness” as a distinct capability node from “trained interoceptive acuity” commits to treating sacrifices for the latter as not-attributable to the former; this is a poset-construction choice, supported by the distinction the framework draws between exercise of a capability and training-level changes to a capability. Under this individuation, the baseline witnesses $R_S \setminus R_B$. An analyst preferring co-extensive individuation would have to look elsewhere for an $R_S \setminus R_B$ witness. Similar candidate witnesses (subject to the same individuation assumption) include baseline peripheral sensory integration and baseline attentional stability.

(d) $R_B \cap R_S$. Consider “private contemplative experience with no material footprint, exercised during time intervals already otherwise-priced by the framework’s poset construction (so that no incremental sacrifice attaches to the contemplation itself).” This is identity-individuated (non-communicable phenomenology, so $\in R_B$) and emits no sacrifice event under the assumed individuation: no money, no time newly displaced from sacrifice-attributable activity, no counterparty, so $\in R_S$. The witness depends on the poset-construction choice that contemplation does not incrementally displace any sacrifice-attributable activity; under a stricter reading that attributes opportunity-cost even to “idle” time, the witness fails. This makes (d) conditional in the same sense as (c). The proposition’s downstream uses (Section 3’s residual cell) only require R_S to be *constructible*, not non-empty; $R_S = \emptyset$ would just collapse δ_{residual} to zero. ■

Remark 24 (The observation-channel carve is the appropriate one for this paper). *The prior residual class was defined by the measurement apparatus (what benchmarks can test). The new residual class is defined by the observation channel (what trades reveal). The observation-channel carve is the appropriate one for a paper about what the framework can observe under its privacy commitments: the question is “what do we actually see?” rather than “what could we test if we had access?” Other carves (benchmarkability, computability) remain appropriate for other questions.*

5.3 Bounding the residual class

GFM’s observational-individuation result [Lasser, 2026c, Corollary 2.1] establishes a *structural* floor on vol_P : for any distinguishable agent k , the poset contains an identity capability d_k with $\text{OI}_{\text{floor}}(k) \geq 1$, so removing a_k contracts vol_P by at least $\text{OI}_{\text{floor}}(k)$. The companion paper’s $\text{vol}_R^{\text{exact}}(O)$ diagnostic (the exercise indicator [Lasser, 2026d, Revealed Sacrifice]), by contrast, sums vol_P -weights of capabilities d with $e_t(d) = 1$, where $e_t(d)$ requires a realized cooperative output in $[t - \Delta, t]$ that would be unrealizable if d were removed from O . Transferring the OI floor into $\text{vol}_R^{\text{exact}}(O)$ therefore requires a bridge between the structural vol_P fact and the temporal exercise indicator. We state that bridge explicitly as a hypothesis rather than derive it from Corollary 2.1.

Definition 13 (OI–Exercise Bridge Condition). *An agent k with OI capability $d_k \in O$ satisfies the OI–exercise bridge at time t if some realized cooperative output in $[t - \Delta, t]$ that agent k participated in is identity-individuating in the sense that removing d_k from O would make that output unrealizable as an output attributable to k . Equivalently, $e_t(d_k) = 1$ under the exercise indicator [Lasser, 2026d, Revealed Sacrifice]. Active presence within O that is merely dormant (e.g., k is nominally in the*

perimeter but contributed nothing to realized cooperative outputs during $[t - \Delta, t]$) does not satisfy the bridge.

Proposition 9 (Active-Agent Floor Transfer, conditional on the bridge). *Fix a window $W = [t_0, t_0 + T]$. For an agent k with OI capability $d_k \in O$ satisfying the OI-exercise bridge (Definition 13) at some $t \in W$, d_k is exercised within W and therefore contributes its vol_P weight $w(d_k)$ to $\text{vol}_R^{\text{exact}[W]}(O)$. Assume additionally that $w(d_k) \geq \text{OI}_{\text{floor}}(k)$, where $\text{OI}_{\text{floor}}(k)$ is the elimination floor of [Lasser, 2026c, Corollary 2.1]. This weight premise holds under the binary-default poset construction of Corollary 2.1 ($w(d_k) = \text{OI}_{\text{floor}}(k) = 1$); under non-binary weight schemes it must be checked explicitly because Corollary 2.1 establishes a contraction floor on capability removal, not a direct lower bound on $w(d_k)$. Summing over agents $k \in K$ whose OI capabilities $\{d_k\}$ are pairwise poset-independent (stated as an explicit hypothesis on the agent set; the category partition of [Lasser, 2026d, Revealed Sacrifice] covers the unbenchmarked space U , not necessarily the observation perimeter O , so this independence is imposed directly rather than imported) and who each satisfy the bridge at some $t \in W$:*

$$\text{vol}_R^{\text{exact}[W]}(O) \geq \sum_{k \in K} w(d_k) \geq \sum_{k \in K} \text{OI}_{\text{floor}}(k).$$

The floor transfers only under both the bridge hypothesis and the weight premise $w(d_k) \geq \text{OI}_{\text{floor}}(k)$; it is not a consequence of Corollary 2.1 alone, which establishes a vol_P elimination floor rather than an exercise witness or a direct weight lower bound.

Proof. By the bridge (Definition 13) at some $t \in W$, $e_t(d_k) = 1$, so $d_k \in P_t^{\text{ex}}$ for that t . Under the window-active form of the companion paper’s vol_R definition [Lasser, 2026d, Revealed Sacrifice], $d_k \in P_W^{\text{ex}} = \bigcup_{s \in W} P_s^{\text{ex}}$, hence d_k contributes its vol_P weight $w(d_k)$ to $\text{vol}_R^{\text{exact}[W]}(O) = \text{vol}_P(O \cap P_W^{\text{ex}})$ (the window-active sum is vol_P on the union of instantaneous exercised subposets, so an exercised capability contributes its full w). By the weight premise of the proposition, $w(d_k) \geq \text{OI}_{\text{floor}}(k)$; combining, d_k contributes at least $\text{OI}_{\text{floor}}(k)$ to $\text{vol}_R^{\text{exact}[W]}(O)$. By the Horizon-Aware paper’s [Lasser, 2026c, Corollary 2.1], $\text{OI}_{\text{floor}}(k) \geq 1$ (the binary default, strengthened by any subsumption weight in vol_P). Pairwise poset-independence of the $\{d_k\}_{k \in K}$ in the parent poset P lets vol_P decompose additively by axiom M4 [Lasser, 2026f, Proposition 1], so the vol_P contributions sum without overlap, giving $\text{vol}_R^{\text{exact}[W]}(O) \geq \sum_{k \in K} w(d_k) \geq \sum_{k \in K} \text{OI}_{\text{floor}}(k)$. Without the bridge hypothesis, Corollary 2.1 [Lasser, 2026c, Horizon Aware] establishes only the structural vol_P -on-elimination fact, which does *not* supply the exercise witness required by the exercise indicator of the companion paper [Lasser, 2026d, Revealed Sacrifice]. Without the weight premise $w(d_k) \geq \text{OI}_{\text{floor}}(k)$, the proof reaches $\text{vol}_R^{\text{exact}[W]}(O) \geq \sum_k w(d_k)$ from the bridge alone, but the further inequality $\sum_k w(d_k) \geq \sum_k \text{OI}_{\text{floor}}(k)$ is left unestablished. ■

Remark 25 (When the bridge holds and when it fails). *The bridge holds when active participation is identity-coupled: the agent’s actions produce realized cooperative outputs whose attribution depends on the agent remaining in the perimeter. Typical operational examples are participation in a committed trade event, execution of a verified action with agent-attributable signing, or contribution to a cooperative capability whose output is signed by the agent’s identifier. The bridge fails for dormant presence: an agent is nominally within O but contributed nothing attributable to them in $[t - \Delta, t]$. In such windows the structural vol_P floor of Corollary 2.1 still holds (removing the agent would still contract vol_P), but $\text{vol}_R^{\text{exact}}(O)$ receives no contribution from d_k until the agent resumes identity-coupled activity. This is consistent with the paper’s stance that vol_R measures what the collective does, not merely what it possesses.*

Remark 26 (Channel separation). *Under the boundary-closed hypothesis on O (the exercise indicator [Lasser, 2026d, Revealed Sacrifice], conditions BC1/BC2), the OI floor contributes exclusively to $\text{vol}_R^{\text{exact}}(O)$ in the combined diagnostic (the exercise-indicator appendix of [Lasser, 2026d, Revealed Sacrifice]):*

$$\text{vol}_R^{\text{combined}} = \text{vol}_R^{\text{exact}}(O) + \text{vol}_R^{\text{lower}}(P \setminus O).$$

It does not contribute to $\text{vol}_R^{\text{lower}}$ (the sacrifice-based bound outside O). Conflating the two channels would double-count. When boundary-closure fails (either BC1 or BC2), the compositional lower-bound interpretation of $\text{vol}_R^{\text{combined}}$ is not established—per the boundary-cooperative-undercount

remark in [Lasser, 2026d, Revealed Sacrifice], $\text{vol}_R^{\text{combined}} \leq \text{vol}_R(P)$ holds only under BC1 + BC2; under BC1 failure $\text{vol}_R^{\text{combined}}$ undercounts boundary cooperatives, under BC2 failure the sign of the gap is undetermined. In either failure regime $\text{vol}_R^{\text{combined}}$ can still be computed and reported as a heuristic composite of the two sub-diagnostics but should not be read as a certified lower bound on $\text{vol}_R(P)$. The OI floor’s contribution to $\text{vol}_R^{\text{exact}}(O)$ is unaffected by either failure mode because the OI capability d_k is, by construction, an internal capability of agent k ’s substrate—not a cooperative or subsumption edge spanning the perimeter boundary—so the OI floor remains a valid lower bound on $\text{vol}_R(O)$ regardless of BC1/BC2.

6 Worked Example: The Dormant-Capability Scenario Re-Staged

6.1 Setup

Two-substrate collective (biology + silicon):

- 8 individual capabilities: 3 biological (b_1, b_2, b_3), 5 silicon ($s_1, \dots, s_5$).
- 8 cooperative capabilities: $\{b_1, s_1\}, \{b_1, s_2\}, \{b_2, s_3\}, \{b_2, s_4\}, \{b_3, s_5\}, \{s_1, s_2\}, \{b_1, b_2\}, \{s_3, s_4\}$.
- 3 private contemplative capabilities (p_1, p_2, p_3) in the residual class R_S —purely private exercise, no sacrifice-observable path.
- Weights: $w(b_i) = 2.0, w(s_j) = 1.0, w(\text{cooperative}) = 1.0, w(p_k) = 1.0$.
- Total $\text{vol}_P = 3 \times 2 + 5 \times 1 + 8 \times 1 + 3 \times 1 = 22.0$.

6.2 Phase 1: Healthy trade (β^{lower} near 1)

The collective operates with diverse trade patterns. Biological agents sacrifice purchasing power and time for bundles that exercise biological capabilities: creative supplies, relationship-building time, embodied experiences. Silicon agents sacrifice compute resources for unbenchmarked optimization targets.

Trade data. 50 above-sufficiency sacrifice events over the window, covering all 16 non-residual capabilities. We adopt the *saturation stylization*: for each non-residual capability c , the bundles and Part-B decomposition coefficients are such that the per-capability max-attribution $\text{vol}_R^{\text{lower}}(c) = \max_{n: c \in Y_n} \alpha_{n,c} \cdot \Delta \text{vol}_P(X_n)$ equals c ’s full vol_P weight $w(c)$. This is a stylized condition for the worked example—coverage alone does not imply saturation, so the equality $\text{vol}_R^{\text{lower}} = \sum_{c \in U} w(c) = 19$ is an example-specific assumption rather than a theorem consequence. Additionally, 12 below-sufficiency events are observed (biological agents purchasing healthcare, food, shelter) but these fail S1 by Definition 6 (under the example’s stipulation that retention is degrading on the relevant need dimension and no third non-degrading alternative is available in the agent’s choice set for each such event) and do not enter the sacrifice aggregate.

Diagnostics. Under the saturation stylization above, $\text{vol}_R^{\text{lower}} = 19.0$ and $\beta^{\text{lower}} = 19.0/22.0 = 0.864$. Gap decomposition under the default *boundary-residual closure* (strategy (c) of Remark 13, which preserves seed-cell attributions cleanly): $\delta_{\text{restricted}} = 0, \delta_{\text{covered}} = 0, \delta_{\text{dormant}} = 0, \delta_{\text{residual}} = 3.0/22.0 = 0.136, \delta_{\text{boundary}} = 0$ (Phase 1 has no cross-cell cooperatives because all non-residual capabilities are covered). No alarm.

Need-sufficiency diagnostic. $\text{NAC} = 3.5$ (an upper bound on true need-access costs per Remark 16). NSE : all biological needs at $\text{sufficiency_status} = \text{met}$, $\text{downstream_safety} = \text{safe}$. Combined diagnostic: ($\text{vol}_R^{\text{lower}} = 19.0, \text{NAC} = 3.5, \text{NSE} = \text{healthy}$). The collective exercises rich above-sufficiency capabilities and needs are cheap to meet.

6.3 Phase 2: Automation concentrates trade (β^{lower} drops)

Silicon capabilities automate tasks previously requiring biological participation. The automation changes *trade patterns*: the biological agents no longer sacrifice time for activities involving b_2 and b_3 .

Trade data. Trade events for b_2 -related bundles: $8 \rightarrow 0$. Trade events for b_3 -related bundles: $6 \rightarrow 0$. Silicon trade events increase. New aggregate: 52 above-sufficiency events covering only 10 of 16 non-residual capabilities.

Cell membership under boundary-residual closure. Among the 8 cooperatives, four are wholly within the covered cell ($\{b_1, s_1\}, \{b_1, s_2\}, \{s_1, s_2\}, \{s_3, s_4\}$, total weight 4) and four cross dormant and covered cells ($\{b_2, s_3\}, \{b_2, s_4\}, \{b_3, s_5\}, \{b_1, b_2\}$, total weight 4). Under *boundary-residual closure*, cross-cell cooperatives are collected into a fifth δ_{boundary} bucket rather than absorbed into either the covered or dormant cell. The closure-adjusted cells are then: $\text{vol}_P(R) = 0$; $\text{vol}_P(C) = w(b_1) + 5 \cdot w(s_j) + 4 \text{ same-cell coops} = 2 + 5 + 4 = 11$; $\text{vol}_P(D) = w(b_2) + w(b_3) = 4$; $\text{vol}_P(S) = 3$ (residual class); $\text{vol}_P(\text{boundary}) = 4$ (cross-cell coops). Total: $11 + 4 + 3 + 4 = 22$.

Diagnostics. Assume *partial* saturation on the covered cell: each covered atomic capability and most covered cooperatives saturate (per-capability $\text{vol}_R^{\text{lower}}$ equals vol_P -weight), *except* one covered same-cell cooperative whose participating events fall short of saturating its weight (e.g., $\{s_3, s_4\}$ with mixed attribution). This gives $\text{vol}_R^{\text{lower}}(C) = \text{vol}_P(C) - 1 = 10$ under the explicit unit-shortfall on the one cooperative. Then $\beta^{\text{lower}} = 10.0/22.0 = 0.455$. **ALARM fires** ($\beta^{\text{lower}} < 0.8$). Gap decomposition (boundary-residual closure):

$$\begin{aligned}\delta_{\text{restricted}} &= 0, \\ \delta_{\text{covered}} &= [\text{vol}_P(C) - \text{vol}_R^{\text{lower}}(C)]/22 = 1.0/22 \approx 0.045, \\ \delta_{\text{dormant}} &= \text{vol}_P(D)/22 = 4.0/22 \approx 0.182 \quad (b_2, b_3 \text{ dormant}), \\ \delta_{\text{residual}} &= \text{vol}_P(S)/22 = 3.0/22 \approx 0.136, \\ \delta_{\text{boundary}} &= \text{vol}_P(\text{boundary})/22 = 4.0/22 \approx 0.182 \quad (\text{four cross-cell coops}).\end{aligned}$$

Check: $\beta^{\text{lower}} + \delta_{\text{covered}} + \delta_{\text{dormant}} + \delta_{\text{residual}} + \delta_{\text{boundary}} = 0.455 + 0.045 + 0.182 + 0.136 + 0.182 = 1.000$.

Closure-policy interpretation. Boundary-residual closure isolates the cross-cell cooperative mass in δ_{boundary} rather than absorbing it into the covered cell (priority-extension, strategy (a)) or sub-partitioning by connected component (strategy (b)). The choice matters for diagnostic attribution. Under *priority-extension*, walk the combined cooperative graph: b_2 is poset-connected to covered nodes b_1, s_3, s_4 (via $\{b_1, b_2\}, \{b_2, s_3\}, \{b_2, s_4\}$), so the entire $\{b_1, b_2, s_1, s_2, s_3, s_4\}$ -component plus its seven incident cooperatives migrates to the covered cell. Similarly b_3 is poset-connected to covered s_5 via $\{b_3, s_5\}$, so $\{b_3, s_5\}$ plus the cooperative $\{b_3, s_5\}$ migrates to covered. After absorption, no dormant atoms remain: $\text{vol}_P(D) = 0$ and $\text{vol}_P(C) = 19$ (11 covered seed plus 4 from absorbed dormant atomic capabilities b_2, b_3 at weight 2 each plus 4 from the four absorbed boundary cooperatives at weight 1 each: $11 + 4 + 4 = 19$), $\text{vol}_R^{\text{lower}}(C)$ stays at 10 (the absorbed nodes carry no new trade evidence), giving $\delta_{\text{dormant}} = 0$ and $\delta_{\text{covered}} = (19 - 10)/22 = 9/22 \approx 0.409$. The alarm fires identically across closure choices because β^{lower} depends only on $\text{vol}_R^{\text{lower}}/\text{vol}_P(P)$; only the cell attribution changes. We adopt boundary-residual here to preserve the directly-readable dormancy of b_2, b_3 in δ_{dormant} and expose cross-cell cooperative mass as a separate diagnostic bucket; priority-extension would collapse the dormancy into a larger δ_{covered} deficit, requiring the gap-decomp reader to recover dormancy from the δ_{covered} share rather than read it directly.

Need-sufficiency diagnostic. NAC rises from 3.5 to 8.2 as automation displaces biological agents from cooperative outputs, forcing them to spend more on basic need-satisfaction (healthcare costs rise without cooperative support structures). NSE: healthcare need for biological agents shifts from $\text{downstream_safety} = \text{safe}$ to $\text{downstream_safety} = \text{degrading}$ (cooperative pathways that previously contributed to healthcare quality are no longer active).

Concentration diagnostic. Suppose the 52 admissible trade events partition across four public categories: silicon-automation bundles 45, mixed coop-silicon bundles 5, biological-only 1, other 1. With equal-magnitude events the category-flow shares (Proposition 7) are $\tilde{\lambda}_{\text{trade}} = (45/52, 5/52, 1/52, 1/52) \approx (0.865, 0.096, 0.019, 0.019)$, giving

$$\widetilde{\text{HHI}} \approx 0.865^2 + 0.096^2 + 2(0.019)^2 \approx 0.758.$$

The uniform distribution over the four categories gives $\widetilde{\text{HHI}} = 1/4 = 0.25$ (the minimum achievable with non-zero leverage on all four). For a governance threshold $\text{HHI}_{\text{alarm}} = 0.5$ (moderate concentration—twice the uniform value, still well below full concentration $\text{HHI} = 1$), $\widetilde{\text{HHI}} \approx 0.76 > \text{HHI}_{\text{alarm}}$ fires the concentration alarm: observed trade is heavily concentrated on silicon-automation, consistent with (though not a determination of) wireheading via the self-reinforcing automation loop. Under committed-mode operation, this statistic is third-party observable from the public committed-event ledger plus the public S1-admissibility labelling—no access to agent-private bundle interiors is required.

Combined diagnostic: ($\text{vol}_R^{\text{lower}} = 10.0$, $\text{NAC} = 8.2$, $\text{NSE} = \text{degrading on healthcare}$, $\widetilde{\text{HHI}} \approx 0.76$). Three diagnostic facets of the same observed trade stream point the same direction (the diagnostics are not statistically independent—all three are functions of the same trade events—but each surfaces a distinct view): the gap decomposition flags biological capabilities as dormant, the need-sufficiency diagnostic flags healthcare as degrading, and the concentration statistic flags silicon-automation as dominating trade flow. Each surfaces a different facet of the same underlying phenomenon (automation displacing biological participation); no single diagnostic would suffice.

6.4 Phase 3: Diagnosis and recovery

The framework diagnoses: two biological capabilities have zero sacrifice evidence. The alarm identifies the *absence of trade signal for biological-valued bundles* as the cause.

Recovery mechanism. The framework does not *impose* exercise (that would be controlled relaxation [Lasser, 2026h, Controlled Relaxation], for the risk dimension). Instead, the framework surfaces the diagnostic to the collective: “biological capabilities b_2 and b_3 have zero sacrifice evidence; the collective’s vol_P includes these capabilities but no agent is trading for them.” The collective can then:

- (a) Investigate whether the capabilities are genuinely dormant (no one values them anymore) or merely displaced (valued but crowded out by automation).
- (b) If displaced: restructure cooperative arrangements to create trade opportunities.
- (c) If genuinely dormant: accept the lower β^{lower} and potentially revise downward the capabilities’ contribution to vol_P .

6.5 Phase 4: The complementary diagnostic

Controlled relaxation [Lasser, 2026h, Controlled Relaxation] asks: are the restrictions on biological capabilities warranted? The revealed-sacrifice channel asks: are the biological capabilities worth recovering?

If the biological capabilities are under restriction, controlled relaxation tests whether the risk is real. If they are merely dormant (not restricted), the sacrifice channel detects the absence and surfaces it. The two papers cover both failure modes: over-restriction (Lasser, 2026h, Controlled Relaxation) and under-exercise (the present paper).

6.6 Lessons from the example

1. **Automation changes trade patterns, not just exercise patterns.** The sacrifice-based diagnostic detects dormancy through the *absence* of trade signal, not through a counterfactual computation.
2. **The alarm detects the problem without agent-interior access.** The framework observes aggregate trade data plus external attestations (S1 admissibility labels, residual classification, sufficiency thresholds, NSE/NAC inputs); it does not require access to individual agents’ interior valuation states. “Without interior access” refers to the agent-internal-state boundary, not to a fully self-contained input pipeline.
3. **Recovery is cooperative, not coercive.** The framework surfaces the diagnostic; the collective decides what to do.

4. **The dynamics parallel the phase-boundary analysis of [Lasser, 2026e, Phase Redundancy].** Dormancy of biological capabilities in the trade data is the same underlying phenomenon as observation-channel degradation: biological contribution to the collective decreases. The phase-boundary analysis treats the dynamical consequence (world-model degradation); the present paper treats the value consequence (vol_P no longer tracks exercised optionality).

7 Integration: Architecture Complements the Measurement Theorem

7.1 Position in the GFM sequence

The companion paper [Lasser, 2026d, Revealed Sacrifice] contributes the measurement theorem—a constructive lower bound on vol_R from voluntary trade events. This paper contributes the architectural layers that make that theorem *operational*: the filter that keeps its admissibility hypotheses satisfied (need-sufficiency), the decomposition that interprets its output (δ -partition of the gap), the concentration-based diagnostic that catches one specific failure mode (wireheading HHI), and the characterization of what the measurement cannot reach (residual class).

The pair fits the preemptive-restriction criterion [Lasser, 2026c, Proposition 1] as follows. Controlled relaxation [Lasser, 2026h, Controlled Relaxation] tests the risk dimension; revealed sacrifice (companion paper) tests the value dimension. When a restriction decision is pending, controlled relaxation asks *is the feared harm real?* and the measurement theorem asks *is the forgone capability valuable enough to recover?* The present paper adds: *how should we interpret the measurement when it fires an alarm, and what is it structurally unable to see?*

7.2 Connection to the Goodhart mechanism

The need-sufficiency architecture (Section 2) gives a concrete mechanism for the kind of proxy failure studied in the Goodhart literature [Strathern, 1997, Mannheim and Garrabrant, 2019]. A framework that fails to distinguish needs from wants—or fails to invert cost-dimension benchmarks—produces a proxy vol_P that is systematically misaligned with vol_R on need dimensions: rewarding agents for having expensive-to-satisfy needs (high cost = high sacrifice = high “revealed value” under a naive reading) when the correct interpretation is the opposite.

The polarity boundary (Definition 3) is the framework’s prophylactic against this specific Goodhart mechanism. Establishing the boundary and filtering below-sufficiency sacrifices via S1 *before* the sacrifice channel operates prevents need-cost inflation from corrupting the vol_R lower bound. The NAC diagnostic (Definition 9) tracks the excluded need-costs separately, making the Goodhart pressure visible rather than hidden.

7.3 Connection to the capability approach

The below-sufficiency / above-sufficiency partition recovers the distinction between basic capabilities and complex capabilities of Sen [1999] and Nussbaum [2011] within the single-poset treatment. The framework adds three formal instruments the philosophical tradition lacks: (i) the downstream-safety condition (Definition 2), which makes the *satisfaction-pathway* requirement structurally rigorous given a threshold (the threshold itself remains a governance input per Section 2.5); (ii) the polarity-correct benchmark discipline (Definition 4), which aligns vol_P -maximization with the agent’s interest on both sides of the threshold; and (iii) the revealed-sacrifice transition (mechanism (c), Section 2.5), which helps estimate the sufficiency threshold empirically as a heuristic input to governance (the mechanism has a bootstrap dependency on prior threshold classification and suggests a transition region rather than uniquely identifying a threshold).

A fuller treatment of the correspondence between GFM and the Sen–Nussbaum capability approach, including a formal fungibility-collapse theorem reducing GFM to standard microeconomics as a special case, is the subject of a companion paper in preparation.

7.4 The shared privacy discipline

The present paper inherits the privacy discipline established in the companion paper and the broader GFM sequence: *resolution through frequency, not through depth*. The gap-decomposition computability (Proposition 4) operates in $O(|P| + E + M)$ using observed trade data plus governance-supplied structural inputs (capability census, poset structure, restriction set, residual class, event-admissibility labels, and a closure policy). These structural inputs are not derived from individual-agent interior state; they are institutional/governance artefacts already needed to operate the preemptive-restriction criterion [Lasser, 2026c, Proposition 1] and the need-sufficiency architecture of Section 2. The wireheading-consistent Part-A category-flow signal (Proposition 7) is separately third-party observable from public committed trade data plus the public S1-admissibility labelling, without requiring the remaining governance inputs listed above. Neither diagnostic requires deeper access to individual agent *interior* state than the companion paper’s measurement channel already permits, but each requires non-trivial external inputs: the gap decomposition consumes admissibility labels (S1 attestation, choice-set facts), restriction set Ξ , residual classification, and governance-supplied sufficiency thresholds; the wireheading concentration signal consumes only trade-flow data plus a governance-set $\text{HHI}_{\text{alarm}}$. “No deeper access to individual agent state” refers to the framework’s agent-internal-state boundary, not to a fully self-contained pipeline.

8 Discussion and Open Questions

8.1 What this paper establishes

Need-sufficiency resolves one polarity-problem sub-case. The sufficiency threshold identifies the structural boundary where sacrifice polarity flips. Below-sufficiency sacrifices in which retention is degrading on at least one need dimension *and* no third non-degrading alternative exists in the agent’s choice set are involuntary and correctly filtered by S1 (Definition 6); below-sufficiency targeting where a non-degrading third alternative is available requires institutional attestation to classify (such targeting may still satisfy S1 because the agent had a non-degrading option). The need-access cost diagnostic (Definitions 9–10) tracks need-satisfaction efficiency as a companion metric.

The B-to-C gap decomposes into disjoint diagnostic cells (4 or 5, depending on closure strategy). Under cooperative-closure semantics (strategies (a)/(b), Remark 13), $1 - \beta^{\text{lower}} = \delta_{\text{restricted}} + \delta_{\text{covered}} + \delta_{\text{dormant}} + \delta_{\text{residual}}$ (Definition 8). Under literal-partition semantics (strategy (c), used in the worked example), the identity adds a boundary bucket δ_{boundary} for cross-cell cooperatives. The cell classification is computable in near-linear $\tilde{O}(|P| + E + M)$ time (Proposition 4) given the capability census, poset structure, restriction and residual sets, event admissibility labelling, and a closure policy. Alarm firings can be interpreted diagnostically by which δ dominates.

A wireheading-consistent signal is computable from trade data. Trade-flow concentration (Proposition 6) provides a tractable concentration signal—Schur-convex in the majorization order and monotone in the concentration of observed trade flow—that is *consistent with* wireheading but does not constitute a wireheading determination on its own. The Part-A category-flow variant is third-party observable from public committed sacrifice events plus the public S1-admissibility labelling that Proposition 7’s formula requires (the labels filter the events the HHI sums over; without them the third-party can compute a flow-concentration score over all committed events but cannot certify it as the S1-admissible aggregate that the proposition characterizes). Interpreting an $\text{HHI}_{\text{alarm}}$ crossing as wireheading requires additional context (governance-set threshold, domain baseline concentration, and cross-diagnostic confirmation via the gap decomposition and need-sufficiency metrics).

The residual class is structurally distinct. Capabilities in R_S (Definition 12) have zero sacrifice evidence by construction and cannot gain it regardless of observation time. When R_S is poset-isolated from the non-residual cells (the residual-cell irreducibility hygiene check of Remark 12), the residual class is the irreducible component of the gap decomposition under priority-extension closure; without isolation, residual mass can migrate to neighbouring cells under priority-extension and the boundary-residual or sub-partition variants of Remark 13 are needed to preserve attribution.

The revealed-observation curve of R_S (as the set of capabilities *structurally outside the channel's reach*) is the appropriate curve for sacrifice-based diagnostics: it is differently shaped from the classical benchmarkability-based curve and, by our interpretation, easier to apply within the framework's observation pipeline. Whether this constitutes “operationally cleaner” in some absolute sense is an interpretation, not a theorem.

8.2 Limitations

Threshold governance is not framework-derived. The sufficiency thresholds s_k and the admissible reference set Π_N^{ref} (Definition 2) are governance inputs, not framework-computed. The framework reads them at evaluation time and does not attempt to derive them from first principles.

The filter's interaction with capability-stuffing. Proposition 3 establishes a pointwise below-sufficiency cap. This is *not* an aggregate dimension-level cap: distinct capabilities each satisfying sufficiency on the same dimension each score 1 and accumulate under M5. Full structural resolution of capability-stuffing requires machinery outside this paper's scope.

The gap decomposition's additivity license depends on the closure strategy. Under the default literal-partition strategy (c) of Remark 13, the exact identity $1 - \beta^{\text{lower}} = \sum \delta$ over the five cells $\{R, C, D, S, B\}$ follows from the strategy-(c) atomic-weight accounting convention $\text{vol}_P(X) := \sum_{c \in X} w(c)$ summed over the disjoint cells (distinct from the inherited M1–M6 poset-measure semantics on poset-independent subsets), with no poset-boundary closure required. Under the cooperative-closure strategies (a)/(b), the four-cell identity over $\{R, C, D, S\}$ holds via M4 only when the cells satisfy CC1 (cooperative-closure) and CC2 (subsumption-closure); priority-extension and global-component sub-partition are the two operational routes to that closure.

The residual class is characterized but not valued. The residual class (Section 5) characterizes *which* capabilities the framework cannot reach, not *how* to value them.

8.3 Open questions

1. **Threshold governance.** Several interrelated questions surround the sufficiency thresholds. (a) *Sensitivity*: if the threshold is set too low, need-costs between the threshold and true sufficiency are misclassified as want-pursuit events; if too high, the framework over-constrains the need. (b) *Dynamic update*: thresholds may shift as technology and social context evolve; a mechanism for threshold revision (analogous to controlled relaxation applied to sufficiency standards) is needed. (c) *Cross-cultural variation*: different populations may set thresholds differently; can the framework maintain culturally-indexed thresholds while preserving a coherent aggregate diagnostic, or does variation fragment it?
2. **Need identification and M5 interaction.** How does the framework identify which capabilities are needs versus wants? The poset-structural approach (high fan-out = need) is a heuristic, not infallible; a hybrid of structural position and revealed-sacrifice polarity transition is likely required. Relatedly, under the capped-benchmark discipline (Definition 4), sufficiency-met needs contribute a fixed cap whose weight is a design parameter affecting the relative importance of need-satisfaction versus want-pursuit in vol_P .
3. **Capability-stuffing as structural avoidance.** Capability-stuffing (Section 2.4) remains an open M5 pathology under this paper's machinery. No signature over post-declaration poset structure or exercise pattern separates stuffed from legitimate capabilities in a way the agent cannot adversarially manipulate. The canonical example reveals itself as an instance of *structural avoidance* flagged as an open gap in [Lasser, 2026f, Discussion]: the agent declines to discover that “travel from home to gas station” and “travel from work to gas station” admit merger into a canonical multi-origin fuel-access capability. Related avoidance pathologies include provenance-free declaration, the Doll Problem (declining exercise), and wireheading (declining broad engagement); these share the structural-avoidance shape but differ in formal detail.

The intended resolution is a *proof-burden taxonomy* with claim-and-challenge back-pressure. Three burden types: proof-of-non-mergeability (stuffing-specific), proof-of-provenance (blocking pure-label declarations), and proof-of-use (blocking persistently-dormant declarations outside the residual class). Proof-burdens operate as canonicalization

preprocessing on P before M5 is applied; M5’s formula is preserved, but the preprocessing introduces substantive new axiomatic content. Claims are temporal and contestable: a prior proof-of-non-mergeability can be voided by subsequent discovery of a merger mechanism, with the discoverer receiving structural-discovery credit per [Lasser, 2026c, Proposition 2] and the avoider losing the M5 credit the voided proof supported. This makes structural avoidance economically risky rather than economically stable.

The sequence provides substrate but not implementation. Exogenous verification establishes commitment infrastructure; the companion paper [Lasser, 2026d, Revealed Sacrifice] provides partial use-evidence via the sacrifice aggregate (incomplete, because the persistent-dormancy attack and private time-sacrifice invisibility remain); The risk controlled-relaxation provides a pattern for adjudication. A dedicated follow-up paper must deliver: the formal canonicalization operator, the three proof-burden types with admissible evidence classes, the claim-and-challenge protocol with challenge-market robustness (admission filtering against spam, identity weighting against Sybil flooding, independence requirements against cartel suppression, bounds on governance capture of the adjudication layer), the discovery-reward calibration integrating with the horizon-aware structural-discovery result, and the structural-avoidance dominance theorem under explicit quantitative conditions (minimum challenger arrival rate, minimum reward-to-cost ratio, maximum adjudication lag, challenger independence). Honest residuals include information-asymmetry persistence, private time-sacrifice invisibility inherited from the companion paper’s P5 scope, collusive-subsidy and external-funding edge cases, canonicalization order/non-uniqueness, and the deceptive-alignment residual that proof-burdens cannot currently address without interpretability. This is substantial invention, grounded in existing sequence substrate but not reducible to it; the natural venue is a dedicated follow-up paper on the structural-avoidance architecture.

4. **Automated verification of poset-boundary closure.** The cooperative-closure strategies (Remark 13(a)/(b)) produce cooperative- and subsumption-closed cells by construction, but verifying closure on adversarially-structured posets may be non-trivial. A structural-verification protocol (analogous to controlled-relaxation testing against poset claims) is a natural follow-up. Under the default literal-partition strategy (c), no closure verification is required since the identity follows from atomic-weight accounting over disjoint cells alone.
5. **OI-exercise bridge conditions.** Proposition 9 transfers GFM’s observational-individuation floor to $\text{vol}_R^{\text{exact}[W]}(O)$ conditionally on the OI-exercise bridge (Definition 13). Where the bridge fails (dormant presence), the structural vol_P floor still holds but $\text{vol}_R^{\text{exact}}$ receives no contribution. A cleaner characterization of the bridge’s domain of validity, and conditions under which the horizon-aware floor transfers directly without the bridge, is open.
6. **Institutional-attestation protocol formalization.** The architecture’s non-ZK-verifiable inputs—sufficiency thresholds, downstream-safety adjudication, residual-class classification, S1 admissibility labels, and the companion paper’s S0/S4(a) and genuine-exchange attestations [Lasser, 2026d, Revealed Sacrifice, Remark on external attestation]—all route through an institutional-attestation layer that this pair specifies in input/output terms but does not formalize as a protocol. A dedicated treatment of the attestation protocol (credentialing, cross-checking, audit cadence, dispute-resolution, and adversarial robustness) is the natural follow-up to this pair and the primary open dependency for deployment. It is out of scope here because formalizing it requires substrate-specific governance design rather than the measurement-theoretic and architectural machinery these papers develop.

Author Contributions

Teague Lasser conceived the need-sufficiency architecture as a poset-construction precondition and the polarity-boundary mechanism that connects below-sufficiency sacrifice to S1 failure. Directed the formal development of the parallel need-sufficiency diagnostic, the gap decomposition with priority-partition semantics, the wireheading-consistent concentration signal, and the revealed-observation curve of the residual class. Identified the shared privacy discipline (resolution through frequency, not depth) that ties this paper’s diagnostic architecture to the companion paper’s measure-

ment theorem. Made final editorial decisions and is responsible for the paper’s intellectual direction and all claims made.

Claude Opus 4.7 (Anthropic) drafted the formal exposition: the need-sufficiency definitions and propositions (Section 2), the gap decomposition and need-access cost diagnostic (Section 3), the wireheading detection via trade-flow concentration (Section 4), the residual class re-characterization (Section 5), and the four-phase worked example (Section 6).

GPT 5.5 (OpenAI) served as cold technical reviewer across thirty review cycles applied to the full pre-split manuscript, catching formal gaps in all sections. Review findings directly shaping this paper include: the explicit priority-filtered partition cells in Definition 8; the full poset-boundary closure (cooperative + subsumption) condition required by the decomposition; the Schur-convexity framing of the wireheading HHI; the existential-quantifier scope of the non-fungibility residuals; the OI-exercise bridge hypothesis for the $\text{vol}_R^{\text{exact}}$ contribution; and the pointwise (rather than aggregate) scope of the below-sufficiency saturation proposition.

Transparency note. Both AI systems operated as tools under human direction. Neither system has continuity across sessions, cannot take responsibility for the work in the sense required by most venue authorship policies, and cannot respond to reviewer queries independently. They are listed as authors to accurately represent their contributions to the intellectual content of the paper, not to claim that they meet all criteria of traditional academic authorship. The corresponding author for all inquiries is Teague Lasser.

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A Proof Sketches and Technical Notes

A.1 Need-sufficiency architecture (Section 2)

Proposition 3 (Below-Sufficiency Saturation, pointwise). Below-sufficiency benchmark for dimension d_k is $\min(\text{actual}_k/s_k, 1)$, bounded above by 1 pointwise. This is a per-capability, per-dimension cap. Distinct capabilities each scoring 1 on the same dimension accumulate weight under M5; the broader class of capability-stuffing via distinct-capability re-instantiation is not blocked by this proposition and is acknowledged in Open Question 3.

Definition 6 (S1-Sufficiency Connection). Below-sufficiency sacrifice events fail S1’s non-degrading-alternative condition of the companion paper [Lasser, 2026d, Revealed Sacrifice] when both hypotheses of the definition hold: (H1) retention degrades the agent below sufficiency on some need dimension d_k , ruling out retention as a non-degrading alternative; (H2) no third non-trade alternative in the agent’s choice set is non-degrading. Together, no *alternative to the trade itself*—neither retention nor any third non-trade option—is non-degrading on every need dimension, which is what S1’s “non-degrading-alternative-to-the-trade” condition requires; S1’s condition fails and the event is filtered. The two hypotheses are jointly required for this structural filter to fire (other S1 failure modes—coercion, duress, information asymmetry—fail S1 via different routes and require institutional attestation rather than this filter): hypothesis (1) without (2) allows S1 to hold via a third alternative; hypothesis (2) without (1) is unmotivated since retention is itself non-degrading.

Proposition 1 (Near-Total Collapse Under Unmet Need). Under homogeneous sufficiency $s_N(c) \equiv s_N$ for all $c \in \text{Down}(p_N)$, $\text{vol}_P^{\text{eff}} \leq g(s_N) \cdot W_{\text{down}}(N) + W_{\text{out}}$ (the upper bound holds without assumptions on other needs because each additional gate factor is ≤ 1 ; full sufficiency of other needs is required only for the equality case). Collapse limits as $s_N \rightarrow 0$ split: zero-floor gates ($\alpha_0 = 0$: linear, ReLU, power-law) contract downstream weight to zero; positive-floor gates ($\alpha_0 > 0$: leaky-ReLU with $\alpha > 0$) contract downstream weight only to the floor $\alpha_0 W_{\text{down}}$.

Proposition 2 (Need-Dominance in the Gradient). Under the common intervention scale (one unit of Δs_N and one unit of Δw have the same cost), the marginal return to closing the sufficiency gap on N exceeds the marginal return to unit-weight addition whenever $g'(s_N) \cdot W_{\text{down}}(N) > 1$. The finite-step form via Taylor expansion (Remark 5) clarifies that this is a per-unit-of-intervention comparison, not a claim about arbitrary-size finite steps.

A.2 Gap decomposition (Section 3)

Definition 8 (Gap Decomposition) and Proposition 4 (Computability). Under the default literal-partition strategy (c), the cells $\{R, C, D, S, B\}$ form a set-partition of P by stipulation (boundary-crossing cooperative nodes go to B rather than to a participant’s cell), and vol_P -additivity follows from the strategy-(c) atomic-weight accounting convention $\text{vol}_P(X) := \sum_{c \in X} w(c)$, which is distinct from the inherited M1–M6 poset-measure semantics on poset-independent subsets: $\text{vol}_P(P) = \sum_{X \in \{R, C, D, S, B\}} \text{vol}_P(X)$ since the cells are set-disjoint and the atomic sum rearranges directly. The Part-B subset-extension of $\text{vol}_R^{\text{lower}}$ via per-capability max-attribution [Lasser, 2026d, Revealed Sacrifice] is similarly set-additive over disjoint cells: $\text{vol}_R^{\text{lower}}(P) = \sum_X \text{vol}_R^{\text{lower}}(X)$ with $\text{vol}_R^{\text{lower}}(D) = \text{vol}_R^{\text{lower}}(S) = 0$ by the dormant/residual definitions and $\text{vol}_R^{\text{lower}}(B)$ a typically-zero contribution from cross-cell cooperatives whose participants span dormant/residual cells (non-zero when admissible bundles cover the cooperative directly; the formula admits either case). Combining, $1 - \beta^{\text{lower}} = \sum_X \delta_X$. Under cooperative-closure strategies (a)/(b), $B = \emptyset$ and the four-cell identity $\text{vol}_P(P) = \text{vol}_P(R) + \text{vol}_P(C) + \text{vol}_P(D) + \text{vol}_P(S)$ requires M4 over poset-boundary-closed cells (CC1: no cooperative spans two cells; CC2: no subsumption edge crosses); the $\text{vol}_R^{\text{lower}}$ step is set-additive in either regime and does not require CC1+CC2.

Computability ($O(|P| + E + M)$) is established by three passes: (1) per-individual-capability priority classification in $O(|P|)$; (2) event-sweep updating dormant $\rightarrow$ covered in $O(E)$ with unconditional $\text{vol}_R^{\text{lower}}$ accumulation; (3) closure-policy pass over the cooperative/subsumption graph in $O(M \cdot \alpha(M))$ for strategies (a)/(b) via union-find, or $O(M)$ outright for strategy (c) via a single hyperedge sweep classifying boundary-crossing cooperatives into B . The Part B admissibility la-

bel must include the full Part B hypothesis set (S0–S5, genuine-exchange conditions, within-bundle poset-independence from the event-local bundle decomposition [Lasser, 2026d, Revealed Sacrifice]).

A.3 Wireheading-consistent concentration signal (Section 4)

Proposition 6 (Concentration Signal). $\text{HHI}(\lambda) = \sum_j \lambda_j^2$ is symmetric and coordinate-convex on the simplex; by the Hardy–Littlewood–Pólya theorem [Hardy et al., 1929, Marshall et al., 2011], any such function is Schur-convex: majorization-ordering of leverage distributions induces monotone HHI ordering. Cauchy–Schwarz bounds the support-normalized minimum $\text{HHI} \geq 1/m$; top- k -mass concentration bounds the extremal side via $\text{HHI} \geq \rho^2/k$. The governance threshold $\text{HHI}_{\text{alarm}}$ is not theorem-derived (Remark 19).

Proposition 7 (Third-Party Observability of the Part-A variant). Let $\mathcal{N}_{S1}$ denote the index set of S1-admissible committed events as marked by the public admissibility labelling. The category-flow statistic $\tilde{\lambda}_{\text{trade}}(j, T) = \sum_{n \in \mathcal{N}_{S1} : \text{category}(Y_n)=j} \Delta \text{vol}_P(X_n) / \sum_{n \in \mathcal{N}_{S1}} \Delta \text{vol}_P(X_n)$ is a direct function of cleartext-committed fields from the committed event definition [Lasser, 2026d, Revealed Sacrifice] together with the public S1-admissibility labelling; no hidden field (agent identity, bundle-interior subset, $\alpha_{n,c}$ coefficients) enters. Diagnostic suppressed when total S1-admissible flow is zero. Per-capability variant requires richer commitments or operation within a consented perimeter O (Remark 21).

A.4 Residual class (Section 5)

Proposition 8 (Residual Class Intersection). $R_B \setminus R_S$ is witnessed unconditionally by (b) artistic practice (identity-individuated, but consumes material sacrifice). Whether $R_S \setminus R_B$ and $R_B \cap R_S$ are non-empty is conditional on poset-construction individuation choices. Conditional witnesses: (c) baseline interoceptive self-awareness (candidate witness under individuation that distinguishes baseline from trained acuity; the strict reading would treat clinical measurement as a sacrifice path attaching to the baseline, in which case the witness fails); (d) private contemplative experience (identity-individuated, no material footprint, conditional on treating idle time as not opportunity-cost-attributable).

Proposition 9 (Active-Agent Floor Transfer, conditional on the bridge). The transfer from the OI result [Lasser, 2026c, Corollary 2.1] to this paper’s $\text{vol}_R^{\text{exact}}(O)$ is *conditional* on the OI-exercise bridge (Definition 13): GFM establishes a structural vol_P floor on elimination, not an exercise witness, so the bridge hypothesis supplies the temporal/semantic conversion to exercise-indicator semantics. Where the bridge holds—identity-coupled participation producing cooperative output within $[t - \Delta, t]$ —the OI floor contributes $\text{OI}_{\text{floor}}(k) \geq 1$ to $\text{vol}_R^{\text{exact}^{[W]}}(O)$ independently of the sacrifice channel. Where it fails (dormant presence), the structural vol_P fact still holds but $\text{vol}_R^{\text{exact}}(O)$ receives no contribution.

B Notation Summary

B.1 Symbols introduced in this paper

Symbol	Definition	Ref.
$\{d_1, \dots, d_m\}$	Dimensions of a need bundle	Def. 1
$\{s_1, \dots, s_m\}$	Sufficiency thresholds per dimension	Def. 1
p_N	Position of need N in the poset	Def. 1
$\text{Down}(p_N)$	Downstream cone of need N	Def. 1
π_N	Need-satisfaction pathway	Def. 1
Π_N^0	Class of threshold-satisfying pathways	Def. 2
Π_N^{ref}	Admissible reference set (governance-designated)	Def. 2
π^*	Element of Π_N^{ref}	Def. 2
$g(s)$	Downstream gate function	Def. 5
α_0	Gate floor ($g(0)$)	Def. 5
$W_{\text{down}}(N), W_{\text{out}}(N)$	Downstream-cone weight, complementary weight	Def. 5
$\text{vol}_P^{\text{eff}}$	Downstream-gated effective volume	Def. 5
Ξ	Restriction set (preemptive-restriction criterion)	Def. 8
β_{alarm}	Alarm threshold	Def. 7
R, C, D, S	Gap-decomposition cells (priority-filtered)	Def. 8
B	Boundary bucket (under closure strategy (c))	Def. 8, Rem. 13
$\delta_{\text{restricted}}, \text{etc.}, \delta_{\text{boundary}}$	Gap decomposition terms	Def. 8
$\text{NAC}(T)$	Need-access cost aggregate	Def. 9
$\text{NSE}(i, N)$	Per-agent need-satisfaction efficiency	Def. 10
$\lambda_{\text{trade}}(j, T)$	Trade-flow leverage (Part B)	Def. 11
$\tilde{\lambda}_{\text{trade}}(j, T)$	Category-flow leverage (Part A)	Prop. 7
$\text{HHI}, \overline{\text{HHI}}$	Herfindahl–Hirschman Index (Part B / Part A)	Prop. 6
$\text{HHI}_{\text{alarm}}$	Wireheading detection threshold	Prop. 6
R_B, R_S	Benchmarkability- and sacrifice-channel-residual classes	Def. 12
$e_t(d), O$	Exercise indicator, observation perimeter	Companion paper
$\text{vol}_R^{\text{exact}}(O), \text{vol}_R^{\text{combined}}$	Exercise-based and combined vol_R diagnostics	Companion paper
$\text{OI}_{\text{floor}}(k)$	Observational individuation floor for agent k	Prop. 9

B.2 Symbols inherited from the companion paper [Lasser, 2026d, Revealed Sacrifice]

Symbol	Definition (companion paper)
(i, X, Y, t)	Revealed-sacrifice event
$\Delta \text{vol}_P(X)$	Surrendered-capability vol_P contribution
$\text{vol}_R^{\text{lower}}, \text{vol}_R^{\text{lower}^A}$	Part B / Part A lower bounds
β^{lower}	B-to-C ratio under revealed sacrifice
S0–S5	Sacrifice-model assumptions
$\alpha_{n,c}$	Event-local decomposition coefficients
(Γ, π, t)	Committed revealed-sacrifice event
$\text{category}(Y), \mathcal{C}$	Bundle’s partition label, public category partition
Commit, U_i	Commitment scheme, agent utility

B.3 Dependencies on prior papers

Paper	Result	Number	Used in
[Lasser, 2026d, Revealed Sacrifice]	Revealed-sacrifice event and assumptions	Def. 1, S0–S5	§2
[Lasser, 2026d, Revealed Sacrifice]	Aggregate B-to-C lower bound	Thm. 2	§3, 4
[Lasser, 2026d, Revealed Sacrifice]	Committed events and ZK rollup	§6	§4
[Lasser, 2026b, Foundation]	Population empowerment	Def. 6	§1
[Lasser, 2026f, Poset Definition]	Benchmark	Def. 2	§2
[Lasser, 2026f, Poset Definition]	Poset measure vol_P , axioms M1–M6	Def. 7, Prop. 1	§3
[Lasser, 2026c, Horizon Aware]	Exercise pathway	Def. 2	Rem. 2
[Lasser, 2026c, Horizon Aware]	Preemptive-restriction criterion	Prop. 1	Def. 8
[Lasser, 2026c, Horizon Aware]	Observational individuation	Cor. 2.1	Prop. 9
[Lasser, 2026c, Horizon Aware]	Information value of structure	Prop. 2	Open Question 3
[Lasser, 2026h, Controlled Relaxation]	Controlled relaxation	Def. 3	§7